

First Light for the GRAVITY+ Adaptive Optics: Extreme Adaptive Optics for the Very Large Telescope Interferometer

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Received Januar 31 (planned), 2025; accepted ...

ABSTRACT

GRAVITY+ is improving by orders of magnitude the sensitivity, sky-coverage and contrast of GRAVITY and the VLTI infrastructure. As a central part of this project, the development of a dedicated high-order and laser-guide star Adaptive Optics (AO) system for VLTI is a crucial aspect. Here we present the design and first results of Gravity Plus Adaptive Optics (GPAO). GPAO is the combination of four state-of-the-art Adaptive Optics (AO) equipping all 8m-class Unit Telescopes (UTs) for the wavefront correction of the VLTI instruments. It offers both Natural Guide Star (NGS, both visible and infrared) and Laser Guide Star (LGS) operations. In this paper, we focus on the design, operations and performances, and first results with this system. We illustrate the improvement brought by GPAO to the interferometric observations with the commissioning done in Fall/Winter 2024. These observations include the first interferometry observations of a redshift $z \sim 4$ quasar, the spectroscopy of a cool brown-dwarf with magnitude $K \sim 21.0$, and first observations of a Class I young star with GRAVITY. The GPAO system will be the workhorse for future high-contrast modes for exoplanets at VLTI and the implementation of the LGS mode within GRAVITY+.

Key words. Instrumentation: adaptive optics – Instrumentation: interferometers – Instrumentation: high angular resolution – Galaxies: quasars: supermassive black hole – Planets: exoplanets

* GRAVITY+ is developed in a collaboration by the Max Planck Institute for extraterrestrial Physics, the Institute National des Sciences de l'Univers du CNRS (INSU) with its institutes LIRA / Paris Observatory-PSL, IPAG / Grenoble Observatory, Lagrange / Côte d'Azur Observatory and CRAL / Lyon Observatory, the Max Planck Institute for Astronomy, the University of Cologne, the CENTRA - Centro de Astrofísica e Gravitação, the University of Southampton, the Katholieke Universiteit Leuven, University College Dublin, Universidad Nacional Autónoma de México and the European Southern Observatory.

1. Introduction

1.1. Optical/IR interferometry with large telescopes and adaptive optics

Adaptive Optics (AO) was identified early on as a prerequisite for interferometry on large telescopes, in order to

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work with apertures larger than the Fried diameter and increase the coherent volume of interferometry. The use of AO directly results in a sensitivity improvement of $\propto D^4$ for diffraction-limited interferometric observations, compared to $\propto D^{1/3}$ in the case of seeing-limited observations (Eisenhauer et al. 2023). As such, AO has rapidly been considered for infrared interferometry to correct the wavefront before interferometric combination in place of multi-speckle interferometry (Roddier 1981), as well as the coupling of light into single-mode fibers (Shaklan & Roddier 1988).

On the Very Large Telescope Interferometer, the role of AO combined to the 8m-class telescopes was emphasized from its initial design: "The largest gains in sensitivity of the VLTI will come from the incorporation of adaptive optics which will make the telescope diffraction limited at $2\mu\text{m}$ " (Beckers 1990). Therefore, a dedicated AO facility was implemented at VLTI from its early start with the Multi-Application Curvature Adaptive Optics (MACAO, (Arsenault et al. 2003)). The emphasis remained on multi-purpose applications, with a trade-off between limiting magnitude and wavefront quality in the absence of laser guide stars. The VLTI was later upgraded with an infrared AO for the Galactic Center, the GRAVITY Coudé Infrared Adaptive Optics (CIAO, (Scheithauer et al. 2016)). On small telescopes, both the 1.8m diameter VLTI/ATs with the NAOMI system (Willez et al. 2019a) and the 1.0 m diameter CHARA telescopes are also equipped with AO (ten Brummelaar et al. 2012).

The next step on large telescopes is the development of an AO facility enabling both Natural Guide Star (NGS) and Laser Guide Star (LGS) operations. Unlike small telescopes, for which the limiting magnitude is set by the number of photons available for fringe-tracking, the limiting magnitude for large telescopes is set by the AO (Eisenhauer et al. 2023). This originates from the fact that the number of photons available for fringe-tracking increases with the collecting area, while for AO this number is set by the number of photons per turbulence cell. For a telescope size larger than the Fried parameter, the limiting magnitude is thus independent of the size of the telescope. This limitation is lifted with LGS AO, where the high-order correction is done on an artificial star. The limiting magnitude in this case is set by the tip-tilt correction, which requires a nearby star in the isokinetic patch. This magnitude is comparable to the fringe-tracking limiting magnitude. Therefore, an LGS-based AO design overcomes the existing trade-off on MACAO between high-order correction on bright targets and sensitive operations on the faint-end. This is the goal of the Gravity Plus Adaptive Optics (GPAO) system, developed in the framework of the GRAVITY+ project.

1.2. GRAVITY overview

The GRAVITY instrument (GRAVITY Collaboration et al. 2017) is a four beam interferometric instrument, offering milli-arcsecond (mas) resolution imaging for objects as faint as $K \sim 21$, together with 30 – 100 micro-arcsecond (μas) narrow-angle astrometry, and microarcsecond spectro-astrometry capabilities in the K-band ($2.0 - 2.4\mu\text{m}$). The instrument allows one to perform phase-referenced observations by using two objects that can be observed simultaneously, one for the Fringe-Tracker (FT) channel which compensates in real-time for the atmospheric piston, and one for Science (SC) channel for long exposures.

GRAVITY has delivered ground-breaking results, from the experimental test of General Relativity in the Galactic Center and the study of the environment of Sgr A* black-hole down to the innermost circular orbit scales (GRAVITY Collaboration et al. 2018a,b, 2019a, 2020a, 2022), to the first spatially resolved observations of a broad-line region in an AGN (GRAVITY Collaboration et al. 2018c), the first detection and characterization of exoplanets using interferometry (GRAVITY Collaboration et al. 2019b, 2020c; Nowak et al. 2020; Lacour et al. 2021), the first resolved image of a gravitational microlens (Dong et al. 2019), and comprehensive spatially resolved observations of disks and the magnetospheric accretion region around young stars (GRAVITY Collaboration et al. 2019c, 2020b, 2021, 2023, 2024b).

1.3. From GRAVITY to GRAVITY+

The goal of the GRAVITY+ project is to increase by several orders of magnitude the sensitivity, sky-coverage and high-contrast capability of GRAVITY and of the infrastructure of VLTI (Gravity+ Collaboration et al. 2022). Since 2021, the phased implementation started with the completion of the upgrade of the GRAVITY metrology, the so-called FAINT mode in order to remove the noise of the metrology laser (Widmann et al. 2022), as well as the upgrade of the global throughput by combining the Beam Compressor and the Differential Delay Lines in the VLTI laboratory (Fabricius et al. 2024), and the upgrade of the fringe-tracker of GRAVITY (Nowak et al. 2024b). These upgrades will soon be complemented by an improved vibration control (Bigioli et al. 2022), and the installation of GRAVITY optics with a high-resolution grism $R \sim 15,000$ and high-precision injection optics for astrometry (Gravity+ Collaboration et al. 2022).

This first phase of GRAVITY+ reached an important milestone with the implementation of the mode GRAVITY Wide (hereafter G-Wide) (GRAVITY+ Collaboration et al. 2022), which allows for the first time wide-field observations with GRAVITY. Before this upgrade, the maximum FT-SC separation was limited to 2 arcsecond on the UTs and 4 arcsecond on the ATs. This requirement is overcome by G-Wide, which enlarges the separation between the FT and SC up to 30 arcsecond. By doing so, it makes possible to pick-up a bright star suitable for fringe-tracking in a much larger area - ultimately limited by the isopiston area - thus increasing the field-of-view and the sample of objects observable with GRAVITY. This upgrade is instrumental for the observations of faint sources with interferometry, in particular extragalactic sources, which are too faint to allow for fringe-tracking. The G-Wide upgrade allowed for the first time to measure a dynamical SMBH mass up to a redshift $z \sim 2$ (Abuter et al. 2024).

The next step of GRAVITY+ is the implementation of state-of-the-art AO. In the current observing scheme, the observations are not constrained by the limiting magnitude on the science object, but by the performances of the AO. On bright objects, this limitation is particularly sensible for high-contrast observations, where a NGS high-order AO correction both improves the Strehl and thus the injected flux in the science fiber, and reduces the speckle noise of the star. This mode is particularly critical for exoplanet observations, for which the MACAO system was not optimized. On faint sources, the limitation of AO arises from the fact that the limiting magnitude of the fringe-tracker star at

magnitude $K \sim 12$ is not bright enough in the visible to allow for AO correction (see Sec 1.1 for the fundamental reason of this limit on large telescopes). With G-Wide, the observations with interferometry are limited to a handful of targets, due to the lack of suitable bright star $V < 14$ in the field of view. The LGS mode will dramatically increase of this sample for the extragalactic case, moving from currently a dozen objects to several thousands.

This paper presents the first interferometric observations with the four GPAO systems installed on the VLTI four 8m telescopes. In the following, we present the Adaptive Optics system (Sect. 2), its operation and performances (Sect. 3), and the first interferometric observations (Sect. 4) which illustrate its improvement on sky. The detailed description of each individual AO subsystem, as well as the analysis and interpretation of the science observations, will be given in forthcoming papers.

2. Instrument design

2.1. General design guidelines and trade-offs

This section summarizes the trade-offs resulting in the final design of the adaptive optics of GRAVITY+. The design of GPAO inherits from:

- ERIS (Davies et al. 2023) for the overall error budget, the implementation of the NGS VIS and LGS WFS, and the RTC frameworks and interfaces,
- AOF/GRAL/GALACSI (Ströbele et al. 2006) for the operational experience at Paranal observatory, especially with the LGS,
- CIAO (Kendrew et al. 2012) for the Infrared WFS hardware and its operation,
- and NAOMI (Willez et al. 2019b) for the overall interface concept with VLTI, as well as the emphasis on automatic and reliable operations.

On the contrary, Appendix A.1 lists the differences between GPAO and these projects that have a significant impact on the design.

2.1.1. ALPAO 43x43 Deformable Mirror

The size of the deformable surface is imposed by the design of the VLTI Coudé train (see Figure 1), where the pupil defined by the secondary mirror (M2) is re-imaged on the mirror number 8 (M8) with a size of 98×101 mm. The voice coil technology from ALPAO¹ was identified as the most cost-effective solution to achieve the required stroke ($> 15 \mu\text{m}$ wavefront-ptp for 3×3 actuators), speed (> 1 kHz) and the necessary pupil size (101 mm).

The actuator density corresponds to an effective pitch of about 20 cm on-sky thus an extreme-AO system, because (1) there is no trade-off between a high order system and a sensitive system thanks to the LGS-assisted mode, (2) there is no trade-off between the number of actuators and the stroke or the speed in the ALPAO technology, and (3) it enables shorter wavelengths which, even if outside the immediate scope of GRAVITY+, are of interest for the long term future of VLTI.

The first DM, delivered in 2022, used a 41×41 pattern with standard pitch 2.5 mm. This DM was used from October 2022 to August 2023 in the GPAO test bench that was

designed and used to perform the AIT of the two complete systems GPAO1 and GPAO2 (Millour et al. 2024, GPAO3 and GPAO4 being tested at subsystems level). Three main conclusions could be drawn from this prototype. First, the ALPAO standard technology (polymer springs) satisfies the requirement in term of actuator gain stability and homogeneity. Considering the risk inherent to any new technology, the mitigation plan of using innovative metallic springs for the next DMs was dropped. Secondly, the speed requirement was reduced from 1000 Hz to 800 Hz (settling time 1 ms to 1.2 ms) to provide more margin for manufacturing uncertainties. Finally, the edges of the pupil were poorly controlled by the 41×41 2.5 mm pattern. The design was thus upgraded to 43×43 with a custom pitch of 2.62 mm. This is the design for the final six GPAO DMs (4 in operations, 2 spares).

2.1.2. OCAM2 camera and frame grabber for the WFS

SPHERE, GALACSI/GRAAL and ERIS use a camera internally developed by ESO based on the EMCCD220 chip from e2v. GPAO instead uses the commercial OCAM2 camera from First Light Imaging² based on the same chip, with comparable performances (sub-electron readout noise, > 1 kHz full frame rate). A major difference is the output in Camera Link Full. First, this video protocol is not supported in the ESO standard. Moreover, the physical distance between the WFS location and the RTC cabinets cannot be crossed by Camera Link standard cables, rendering a media conversion necessary. GPAO therefore uses a commercially available frame grabber from Pleora in order to execute the conversion from Camera Link Full protocol over copper cable towards 10Gb-ethernet e-vision protocol over optical fiber. The first two OCAM2 camera were delivered beginning of 2021, allowing us to test the stability of the data stream and the additional latency in the frame grabber. A total of 12 camera have been procured (4 NGS VIS, 4 LGS, 4 spares).

2.1.3. Shack Hartmanns versus Pyramids

The Pyramid sensor may appear as a more promising solution than a Shack-Hartmann (SH) for GRAVITY+ because of its improved sensitivity to the low orders which are critical to inject inside a single-mode fiber or reach high contrast at a small inner working angle. Moreover, assuming a detector with sub-electron readout noise, a Pyramid sensor allows to reduce the number of measured modes without modifying the optical setup (e.g., when the NGS WFS is used as low-order truth sensor in LGS mode).

For the LGS WFS, the Pyramid sensor has been discarded because the critical low orders are sensed by the NGS sensor anyway and because the behavior of a Pyramid with the elongated spot is not completely understood.

For the NGS VIS WFS, the Pyramid sensor has been discarded for three reasons. First, the large amplitudes NCPA created by the downstream VLTI optical train challenge the linearity of the Pyramid sensor. Second, the selected DM technology suffers from gain instabilities and creep (with typical time scale of several seconds to days). To mitigate the risks on the overall system, we adopted the principle to associate this possibly non-linear actuator with

¹ www.alpao.fr

² www.first-light-imaging.com

a linear, reproducible sensor so that a recalibration is always possible. And third, the development of the SPARTA-upgrade RTC to support the Pyramid with respect to the baseline ERIS algorithms were in contradiction with the "success oriented" schedule of the project. We also considered the lack of operational experience with the Pyramid sensor within ESO, while the SH had been operated with a high level of reliability for many years, which is critical for a complex system like VLTI.

2.1.4. Number of measured modes

The philosophy motivating the design is the following: the GPAO NGS mode is optimized for robust peak performances on bright star, the GPAO LGS mode is optimized for robust performances toward the faint end.

On the one hand, considering the DM characteristics, the maximum useful repetition rate of the AO loop is 2 kHz, which is matched by the OCAM2 camera in full frame. Therefore, there is no trade-off between the number of pixels to read and the speed. On the other hand, the RTC prototyping activity predicted a maximum loop rate of 2 kHz for an ERIS-like system ($\approx 40 \times 40$ sensing, ≈ 1200 actuators). So, there was no trade-off between the number of sub-apertures and the speed. Consequently, the baseline design for the GPAO NGS mode was to control up to 900 modes with a 40×40 SH.

For the LGS WFS, we decided to lower this number to 500 modes with a 30×30 SH. This modification was motivated by the lower transmission at the coudé focus (GPAO) compared to the Cassegrain focus (ERIS). The 30×30 provides improved robustness against situation of low-sodium density. A 20×20 , while obviously even more robust to lower flux return, would have significantly degraded performances in the case of average seeing and hampered the perspective of shorter wavelengths at the VLTI (see for instance Patru et al. 2020).

2.1.5. Piston-free Operation

An adaptive optics system for an interferometer shall provide piston-free operation. There are subtleties in this definition (e.g mean defined over pupil or pupil apodized by the single mode fiber of the instrument) but the analysis is identical at first order. The piston properties of the ALPAO mirror technology were carefully investigated by the NAOMI project (Wollez et al. 2019b). First, this study demonstrated that it is possible to construct a piston-free modal basis, without the need for a position sensor (Le Bouquin et al. 2018). Secondly, this study showed that the stroke of piston-free modes is only slightly reduced compared to their optimal counter-part including piston, and only for low order radially symmetric modes (focus, spherical, and higher order spherical). Altogether, we concluded that GPAO does not have specific requirements on the DM manufacturing or on the design of the RTC.

2.1.6. Integration of CIAO

CIAO is the 9×9 SH WFS providing off-axis IR wavefront sensing for the observations of the Galactic Center with GRAVITY (Scheithauer et al. 2016). It is fed by the second beam getting out of the STS at the coudé focus of the UTs.

The CIAO Real Time Calculator is built on the *SPARTA light* platform. It controls 40 modes, to match the degree of freedom of the MACAO deformable mirror.

Supporting CIAO operation within the context of GPAO is mandatory. We first explored a conservative approach, in which the new ALPAO DM could be exposed as a device fully compatible with the former CIAO RTC (vector of 40 commands in a sFPDP interface). A more ambitious approach was chosen: the CIAO WFS is fully integrated as an additional sensor of the GPAO system. The opto-mechanical hardware and the local controllers remained unmodified. The RTC and high level control hardware and software have been decommissioned.

This choice was motivated for its opportunity of combining the CIAO IR sensor with the new LGS sensor. This is the optimal configuration for the Galactic Center because it allows to use the nearby, IR-bright GCIRS7 star ($K=6.5$ at $6''$) for sensing the tip/tilt/focus modes (goal 40 modes) and to use the LGS for sensing the higher order modes. Without CIAO, the closest visible star is a $V=14$ at $15''$.

2.1.7. Zonal versus Modal control

Fundamentally, the control in GPAO is modal, in the sense that the control space is determined by a subset of pre-defined modes (approximate KL), and not by a subset of the eigenvectors of the WFS+DM system. This classical approach in modern Adaptive Optics delivers improved performances by focusing on the modes with highest turbulence power. The control can be *explicitly modal* (the IIR filters are expressed on modes) or *implicitly modal* (the modal space is defined by the filtering of the reconstruction matrix, and the same IIR filter is applied to all elements of the command vector). Systems exist in both approaches.

The *explicitly modal* approach allows to tune the IIR filters differently on each mode, for instance to reject a known vibration on a given mode, or to implement large leaks on high order modes. But its implementation in SPARTA suffers from an increased latency because of a post-filter matrix multiplication (KL modes to DM commands). Considering the risk on the latency and the necessary development time, it was decided to implement the GPAO high order controllers as *implicitly modal*.

Ultimately, the controllers are still able to perform modal leaks: the last command vector (DM space) is filtered by a $N_{act} \times N_{act}$ matrix, and the result is subtracted from the next delta-command at the input of the controller. The filtering matrix partially leaks the controlled modes depending on their order, and fully leaks the uncontrolled modes. The mechanism is similar to the garbage collector used in SPHERE, but the operation is here executed for each frame in real-time. The projection is executed in the dead time of the RTC, and thus does not increase the overall latency.

2.1.8. Software de-rotations

GPAO is unusual by the number of rotations inside the system. The photometric pupil and its spiders (M2), as well as the Jitter mirror of the laser launch telescope, rotate with respect to the WFS by the Azimuth - Elevation angle. The DM rotates with respect to the WFS systems by the Azimuth angle. We decided to avoid any mechanical dero-

tation inside the WFS, that is to let *both* the photometric pupil and the DM pattern rotate on the WFS. This strategy rigidifies the entire ensemble of DIC + STS + VLTI + WFS into a common, non-rotating frame (see Figure 2). It simplifies significantly the geometrical modeling of the instrument and the management of the NCPA.

The strategy implemented in the former CIAO (measuring the DM/WFS interaction matrix at many Azimuth angles, defining a sub-set of system modes properly seen at all angles, and pre-computing a bank of control matrices) would be extremely time-consuming in calibration considering the number of modes in GPAO. The strategy implemented in NAOMI (rotating the control Zernikes) is not possible in GPAO because the control space (KL modes) does not include the rotation operator. Instead, GPAO relies on pseudo-synthetic Interaction Matrix (PSIM, see Sec. 2.3.4) created with a geometric model of the instrument. The angle between the DM and the WFS is one of the parameter of the geometric model. For GPAO, we dramatically speed-up the computation of the synthetic interaction matrix, so that the interaction matrix can be recomputed on-the-fly based on the actual Azimuth angle. The highest update rates (3 s) matches the requirement to cross the meridian at a zenithal distance of 5 deg while controlling all KL modes.

2.1.9. Patrol fields

For the NGS WFS, it makes sense for the patrol field to be larger than the iso-kinetic angle in the longest wavelength used by VLTI (10 μm), in good seeing conditions. In practice, this means covering the entire 1.5 arcmin (radius) telecentric field of view at the coudé focus of the UTs.

For the LGS WFS, we decided to also patrol the entire coudé field of view. It enables the possibility to put the LGS star on any of the two objects transmitted to VLTI (one being the fringe-tracking star, the other being the science object). It also allows to add small pointing offsets to the LGS spot, fixed in pupil orientation, as a mitigation to potential Rayleigh contamination.

2.2. Sub-systems description

2.2.1. Implementation in the Unit Telescopes

Figure 1 highlights the location of the main subsystems of GPAO within the structure of the Unit Telescope (). Relevant optical elements are the secondary M2 which defines the aperture stop of the telescope, the deformable mirror (DM) located in M8 and conjugated with M2, the dichroic beam splitter M9 which reflects the infrared part $> 1 \mu\text{m}$ for the VLTI/STS/CIAO and transmits the optical light to the visible WFS, and the Telecentric Doublet lens located just below M9 in the transmitted coudé focus. This lens acts as a (quasi) field lens with a clear aperture of 230 mm and provides a telecentric field plane of 1.5 arcmin at f/47, the so called patrol field. The M9 dichroic being in a convergent beam close to focus, it generates field-dependent aberrations and pupil distortions in the transmitted beam.

2.2.2. Wavefront sensors

The new Wavefront Sensors of GPAO enable NGS VIS and LGS operations (Bourdarot et al. 2024), and are installed

at the coudé focus below the UTs. The design of the WFS is based on the AO module of ERS (Riccardi et al. 2018), adapted from the f/13.6 beam for a Cassegrain focus to an f/47 beam. In the coudé, the WFS units are located below the VLTI STS and M9 dichroic (Fig 1). In the WFS unit, the Periscope Beam Splitter (PBS, a plano-parallel dichroic) reflects the light of the sodium laser line around 589.195 nm toward the Laser Guide Star (LGS) WFS, and transmits the broad band natural light (600-1000 nm) toward the Natural Guide Star VIS WFS. The NGS VIS WFS and the LGS WFS are both installed on a translation XY-stage which cover the full patrol field. The radius of the patrol field depends on the AO mode which is chosen, and is indicated in Table 2.3.4, .

NGS VIS WFS The NGS VIS WFS is a Shack-Hartmann sensor which can be configured either as the visible High Order (HO) sensor, or as the visible Low Order (LO) sensor in conjunction with the LGS WFS. The NGS VIS WFS is composed of 8 main subsystems. (i) The *ADC* (Atmospheric Dispersion Compensator) is composed of a pair of Amici prisms, and designed for a maximum zenith angle of 70°. (ii) The *Technical CCD*, uses the blue part of the spectrum $< 400 - 570 \text{ nm}$ ($R > 95\%$) reflected from an internal dichroic (DCR). It allows to image either the pupil or the field using a moving lens, and can be used as an acquisition camera with 20" field of view. (iii) The *PSM* (Pupil Steering Mirror) adjusts the position of the pupil with a piezo-driven mirror, located close to an intermediate image plane with f/20. (iv) The *NDIA* field diaphragm, located in the image plane, (v) The *Filter Wheel* is a set of 1:10 and 1:100 Neutral density, a He-Ne notch filter, a Sodium notch filter, and a long-pass $> 800 \text{ nm}$ filter for best performances on bright stars. (vi) The *NROT* derotator, made of FK5 prism. In normal operation, this device is fixed. (vii) The *HO/LO* translation stage switches between either the 40×40 subapertures (HO) or 4×4 (LO) lenslet arrays. The exit pupil diameter is fixed to 5.75 mm. (viii) The OCAM2 camera measures the output spots, with a full detector size of 240×240 pixels and $24 \times 24 \mu\text{m}$ pixel size. The multiplication gain enabled by the *emCCD* ranges from 1 to 1000. The overall bandpass in NGS VIS mode (600-1000nm), including the coudé train and the NGS WFS sensor, matches the *G_{RP}* bandpass (Gaia Collaboration et al. 2021).

LGS WFS The LGS WFS is a Shack-Hartmann sensor with a fixed configuration. The beam is converted from f/47 to f/12 in order to adjust for the large focusing range. The LGS WFS is composed of 4 main elements. (i) The *LFOC* focusing stage holds all the optics from L3 onwards. It allows to adjust the focus from 80 km to infinity. (ii) The *LROT* derotator is kept fixed in normal operation (cf NGS). (iii) The *LPSM*, located close to the image plane, steers the pupil based on the same principle as in the NGS. (iv) The 30×30 lenslet array is fixed inside the OCAM2 camera. When observing in LGS mode, the NGS and LGS camera are synchronized by a physical link distributing a common clock.

Calibration Unit The calibration unit of the WFS subsystem consists of two multimode fibers (one for the NGS, one for the LGS) located in the respective focal point

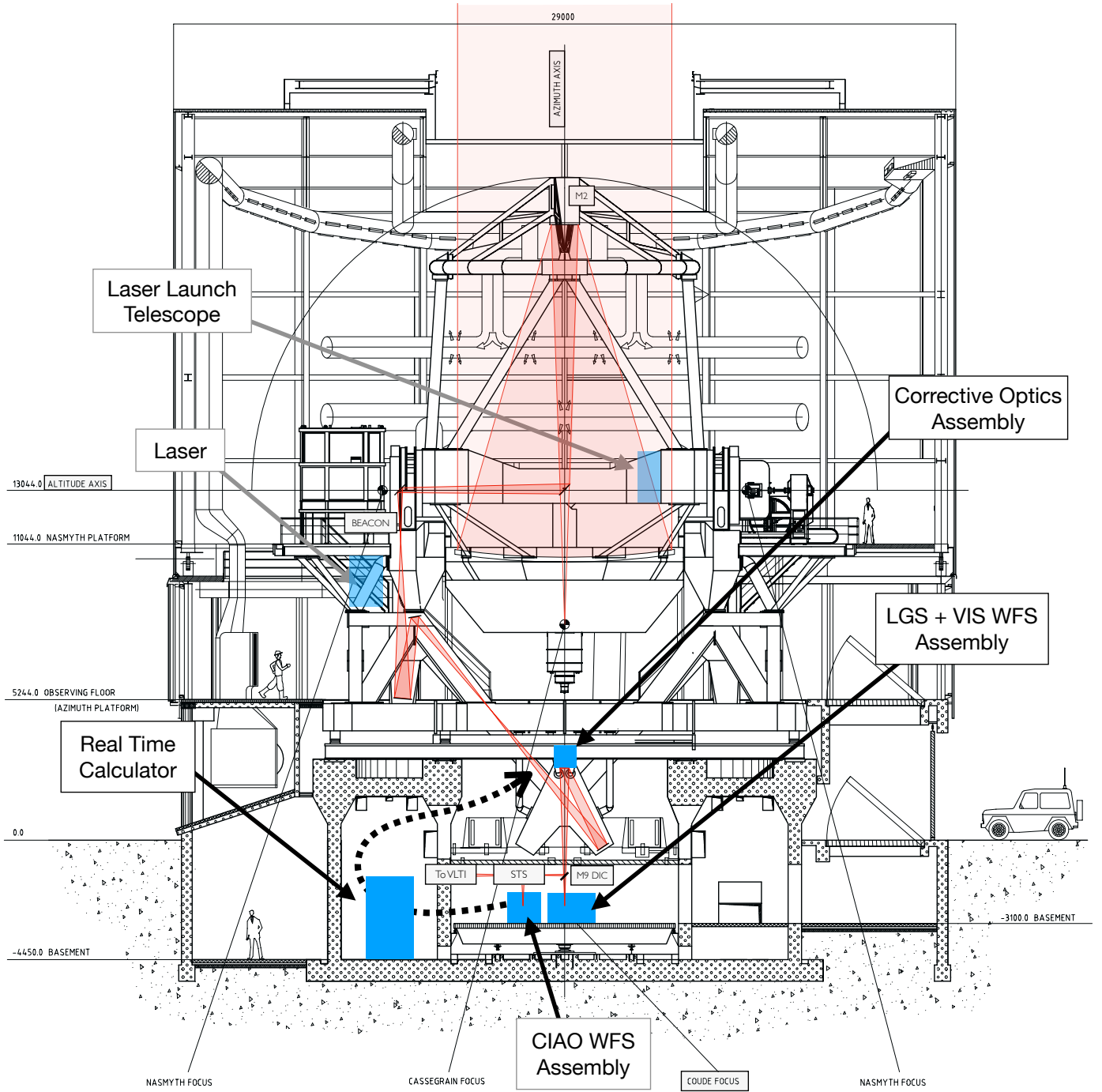


Fig. 1. Location of the GPAO's subsystems in the structure of the Unit Telescope. The location of the M2 mirror, the elevation and azimuth axes, the Nasmyth beacon, the M9 dichroic, the star separator and the path to VLTI are highlighted because of their particular importance for GPAO.

of the WFSs. The fibers are fed with a broadband halogen lamp which covers the optical/NIR spectrum (400-1000 nm). During night-time operation, the calibration unit is moved out of the optical path by a linear translation stage on which the unit is mounted.

2.2.3. Real Time Calculator

The Real Time Calculator is made of two components. First, the Hard RTC (HRTC) is responsible of the main low-latency control loop. Second, the Soft RTC (SRTC) is

in charge of monitoring and driving the HRTC with high level, but not real-time, tasks. GPAO re-uses and improves many features of the ERIS RTC (Baruffolo et al. 2018), the AOF RTC (Kolb et al. 2012) (e.g the real time calculations blocks) and of the NAOMI RTC (e.g the chopping and the saturation management, and the communication with the ALPAO deformable mirrors).

HRTC The HRTC reads the camera pixels, computes a corrective command vector and applies it as fast as possible. It is based on the SPARTA-upgrade platform from ESO

(Shchekaturov 2023), which replaces the SPARTA RTC box by a single workstation. The HRTC is fed by the WFS cameras. The OCAM2 uses camera link connectors, so a Pleora CL-Ten frame grabbers ensures a 10 Gigabit Ethernet (GbE) connexion to the HRTC. The Saphira detector of CIAO uses a Serial Front Panel Data Port (sFPDP) interface, which was ported in the SPARTA upgrade. Once computed, the commands are sent to the ALPAO DM, using the ALPAO HSDL protocol itself on a 1 GbE private network (one per electronic). The HRTC also sends real-time tip/tilt commands to the JM and to the FSM of the LGS launch telescope, in ESO MUDPI format through a 10 GbE link. The hardware chosen for the HRTC is the Dell PowerEdge R7525 server, equipped with additional Intel X710-T4 Network Interface Cards (NIC) and one New Wave DV v5051 sFPDP Card. The system is configured to run as deterministic as possible (diskless, load only required interfaces, confine the kernel). It takes advantage of its multi-CPU architecture, of the principle of Interrupt Request (IRQ) affinity and of the NUMA architecture. The HRTC components use non-blocking communications and shared memory areas to exchange intermediate results so that the data can be processed in parallel to their reception as much as possible.

SRTC The SRTC is the interface between the HRTC and the outside world. It catches the HRTC real-time stream, computes telemetry data, analyzes performance parameters, triggers alarms, and updates the matrices and references vectors. The SRTC mostly interacts with the Observation Software (OS) and its Secondary Loops (SL), to which it sends measurements (e.g pupil position, mean-tip/tilt...), and from which it receives update and commands (e.g new rotation angles, trigger to recompute the control matrix, new NCPA vector...). In terms of hardware, the SRTC is implemented as a standard SPARTA cluster composed of several workstations (concentrators, crunchers and gateway), connected via a 10 GbE switch.

The SRTC implements different modules that were adapted or improved to meet GPAO needs. **(i)** The ‘PSIM module’ implements the synthetic model of the system, and was updated to run faster and to include the model of the ALPAO DM and new parameters to fit the system degree of freedom, see Sec. 2.3.4. **(ii)** The AOF ‘misregistration monitor’ fits the PSIM parameters, either from calibration or inferred from the AO telemetry, as described by Béchet et al. (2012). **(iii)** This module was cloned into a ‘shift monitor’ dedicated to lateral error estimation. It implements innovative methods (Berdeu et al. 2024) with a fast and robust, but perturbative, solution for the automated acquisition, see Sec. 3.1.2, and a non-invasive approach for closed loop operations, see Sec. 2.3.5. **(iv)** The ‘spot monitor’ merges the specific AOF modules dedicated to HO NGS, LO NGS and LGS into a single versatile module. The fitting strategy was changed for a global inverse problem approach that robustly fits the parameters jointly through all the SH-WFS spots: spot flux, spot size in NGS, sodium layer thickness in LGS, elongation and orientation of natural extended objects, and individual or average spot position. The robustness of the approach makes it an essential tool for the automated acquisition, see Sec. 3.1.2 (star detection, star centering, magnitude estimation). This module also computes the weight maps for the weighted center of

gravity used on faint or extended objects. **(v)** The ‘loop monitor’ analyzes the AO loop telemetry (DM commands, slopes and intensities) to compute average offloads (focus, tip-tilt) or to raise alarms (DM clips or saturations, occurrence of OCAM2 random bad columns, under-illumination, over-illumination) to inform the OS which is in charge to open or not the loops (AO and/or SL) and take action. **(vi)** The ‘atmosphere performance monitor’ estimates the atmospheric conditions (r_0 , L_0 , seeing, Strehl, ...) based on the pseudo-open loop slopes computed from the AO telemetry. It is directly inherited from AOF and was just made versatile to the variety of SH-WFS used in GPAO (9×9 , 30×30 , 40×40). A new wind speed estimator has been integrated as well, based on a frozen flow multi-layer model. **(vii)** Finally, the ‘pupil monitor’ follows the lateral position of the photometric pupil. It was fully re-implemented to be versatile to the different SH-WFS geometries of GPAO, to include the M2 shadow (central obscuration and spiders), and with new options to finely fit the pupil magnification and relative angle.

2.2.4. Corrective Optics and Deformable Mirror

The Corrective Optics Assembly (CO) is located in the M8 position of the UT. It is composed of the Quasi Static Mount (QSM) providing static XYZ alignment and motorized Gimbal tip/tilt, and of the ALPAO Deformable Mirror (DM) itself. A CAD view can be found in Figure XXX.

The ALPAO Deformable Mirror features 1432 actuators on a square 2.62 mm pitch pattern. The stroke of pushing a group of 3×3 actuators is typically $20 \mu\text{m}$ ptp-wavefront, with respect to the zero position (that is: this stroke is achievable in both directions). The first resonance is around 1000 Hz. The settling time at 10% is about 1.2 ms, including the drive electronic latency. The DM is controlled by two standard Drive Electronics of ALPAO, each featuring two Ethernet connections. One is dedicated to the house keeping and monitor, and the other to the high speed data link (1 Gb/s). The DM is equipped with cooling plates on two of its lateral faces. The flow of coolant is regulated to maintain the DM internal temperature to within $0, +3$ deg of the ambient temperature at the location of the DM.

The center of the DM (actuator 716, starting 0) is located on the Azimuth axis of the telescope with an accuracy better than 0.5 mm. The DM is mounted in a quasi-static gimbal mount. It uses a basic concept of flex-pivot to provide a large range (2 deg-mechanical, > 2 arcmin-sky), slow, absolute tip/tilt with the vertex in the centre of the DM surface. The actuators M-235.2DG from Physics Instrument are absolutely encoded and provide a tip/tilt resolution of 10 mas-sky. This actuation serves to steer the Nasmyth Beacon anywhere in the coudé field of UT, and to offload the DM tip/tilt when doing internal calibration. When observing on sky, the QSM is kept static at the position corresponding to the optical axis of the telescope.

The CO Proximity Cabinet is located next to the Corrective Optics, inside the inner track of the UT. This was necessary to keep the distance between the DM and its electronic shorter than 5 m. This cabinet embarks the two DM electronic racks (each driving 700 actuators), the Beckhoff controllers of the motors of the gimbal motion, and an Ethernet switch used to convert the incoming high speed and housekeeping networks (transported over fibers) into copper cables.

2.2.5. Laser Guide Stars

GPAO will be operated with the LGS to be installed on UT1-3 (Booth et al. 2024) and any of the four LGS installed on UT4 (Calia et al. 2014). The laser sources of UT1-3 are essentially identical to those used in the UT4 Adaptive Optics Facility (Karpov et al. 2011) and those that will be used on the ELT. The LGS laser source feeds a laser projection subunit (LPS) mounted on the side of the UT primary mirror. The LPS of UT1-3 are identical to those used on the ELT (Nijenhuis et al. 2024) and are comparable in performance to the one of UT4. The LPS is basically a reversed telescope equipped with a slow and a fast steering mirror (LSM and JM) used to control the beam direction. Each UT is equipped with its own camera-based aircraft avoidance system to prevent beam collisions with planes.

2.2.6. Control Software

The control software of GPAO is independent for each telescope, both in hardware and software. Each GPAO has two workstations running the framework of the VLT Software. The GPAO Instrument Workstation (`wgpXao`, with `X` ranging from 1 to 4) hosts an Observation Software (OS), a Technical Detector Control Software (TDCS) and an Instrument Control Software (ICS). A Detector Control Workstation (`wgpXdcs`) acts as command handler layer toward the camera.

The Instrument Control Software (ICS) controls and supervises all GPAO devices, including all devices of the former CIAO instrument. It communicates to the hardware via local controllers: the Beckhoff WFS PLCs (`pgpXics1`), the Beckhoff CO PLCs (`pgpXics2`) and the CIAO LCUs (`lgpXics`). It is the only interface for piloting the motion of axes. It reads the sensor telemetry and house keeping, including those from the DM.

The Observation Software (OS) coordinates the activities of the ICS, the TDCS, and the RTC. It collects the states and logs of all subsystems and constructs the main state and log of GPAO. It executes the acquisition sequence. It runs the secondary loops. It prepares telemetry data to be archived. Toward the outside, it receives commands from the VLTI Supervisor (ISS), it sends commands to the Telescope Control Software (TCS), to the star separator of the VLTI (STS) and to the Multi Laser Guide Star Facility (MLGSF) of the Unit Telescope.

2.3. Control strategy

2.3.1. KL modal basis

The modal basis of GPAO uses approximate Karhunen-Loève modes defined in DM space. The method defines active (controlled) and passive (“slaved”) actuators. The covariance matrix between all actuators is built using their relative distance and the structure function of the turbulence. The eigenmodes of the covariance matrix of active actuators (inside the pupil) form the controlled modes. These modes are extrapolated to the passive actuators (outside the pupil) by projecting their contribution to the controlled modes. In practice, we force the 6 first modes to be exact Zernikes (piston, tip, tilt, focus and the two astigmatism) and remove the piston from the final basis. The controlled pupil is restricted to an annulus extending from 13.95% to

100% of the pupil diameter. This defines 1116 independent modes and 316 passive actuators. The exact same modal basis is used for all WFS in all configurations (NGS_VIS, NGS_IR, LGS_VIS, LGS_IR).

The number of inverted modes (e.g. controlled modes) is changed depending on the WFS in use, when building the control matrix. The values can be found in Table 2.3.4. For the high order natural guide star wavefront sensors, the number of controlled modes is reduced when approaching the limiting magnitude. These numbers have been selected to maximize the robustness of the system while still matching the requirement in terms of Strehl ratio in the K-band under median seeing conditions.

When using the 40×40 WFS, the system has significant margin to control more modes, possibly up to ≈ 950 . This will be explored in the next years, especially in the context of the incoming J-band instruments for the VLTI (Bifrost, Kraus et al. 2022).

2.3.2. NGS/LGS combined control

In NGS mode, all controlled KL modes are sensed by the NGS WFS and corrected by the HRTC with the Deformable Mirror. The situation is more complex in LGS mode. The tip/tilt/focus are measured in the NGS WFS and the higher order modes are measured in the LGS WFS. Both are corrected by the HRTC with the Deformable Mirror.

Because of the up/down propagation through turbulence, and because of the slowly varying distance of the Sodium layer, the above scheme does not control the tip/tilt/focus of the laser spot onto the LGS WFS. Instead, this fast tip/tilt is corrected in real-time by the HRTC with the Jitter Mirror (JM) of the launch telescope. The JM offloads to the Laser Steering Mirror (LSM). The slow focus drift of the laser spot is corrected by a Secondary Loop on the OS: it reads the averaged focus from the SRTC, and acts on the LFOC device accordingly.

2.3.3. Tip/tilt offload

Offloading the tip/tilt modes from the Deformable Mirror reduces the overall stroke budget and cancels the fitting error of the tip/tilt modes. A dynamical Gimbal mount for a DM weighting 5 kg was considered too complex for GPAO. Instead, the initial GPAO design considered offloading the tip/tilt into the M2 itself, using the available “rapid-guiding” link of the VLT.

When configured in rapid-guiding, the M2 delivers tip/tilt residuals of 20 mas-rms. This is satisfactory for seeing-limited observations, but needs to be further reduced for AO-assisted systems. Unfortunately, preliminary tests confirmed that most of the power of this noise is above 100 Hz, exactly in the amplification bandwidth of GPAO (see Fig. E.1). In the meantime, we could verify that the stroke of the DMs alone was sufficient to observe up to a seeing of 1.5". The rapid offload to M2 was therefore discarded and replaced by a slower offload to the telescope axes directly, every 3 s. All the communication and control mechanisms remain in place, so the rapid offload could be revived if the M2 noise is reduced in the future.

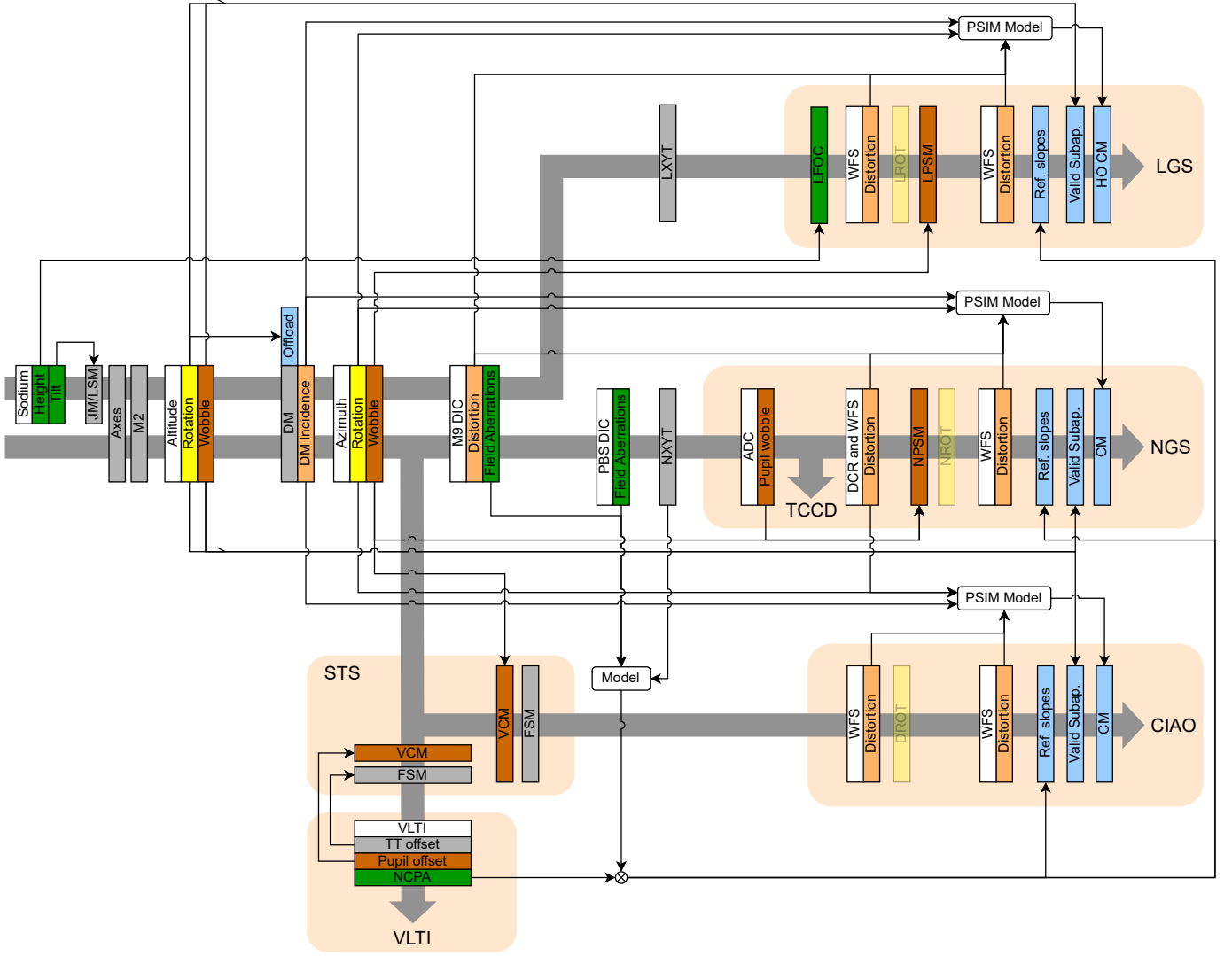


Fig. 2. Control scheme of GPAO. Disturbances and corrections are represented, but sensing is not represented.

2.3.4. PSIM model

The AO control scheme in GPAO is based on pseudo-synthetic interaction matrices (PSIM, Oberti et al. 2006). (I) ‘synthetic’ indicates that the IMs are computed from a pure numerical model of the interaction between the DM and the WFS. (II) ‘pseudo’ emphasizes that this model only depends on a few number of physical parameters that are fitted on the real system.

(I) The numerical model is based on a pure geometrical propagation of the incident wavefront: the WFS slopes are canonically given by the integrated phase differences on the lenslet sides. The DM is modeled by a set of 2D influence functions defined on a high resolution Cartesian grid and which are the same for all actuators. Their 2D model has been fitted on the mean influence function, from the calibrations provided by ALPAO. A non-axisymmetric description of the influence function is necessary to account for the inter-actuator cross-coupling, to avoid introducing artificial fitting error, and to properly match the measured slopes. The actuator pattern and influence functions of the DM are stretched in one direction of the actuator pattern to model the incidence of the beam and the DM tilt in its

Corrective Optics mount. We ignore any other distortion terms of the actuator pattern.

(II) The PSIM model includes a limited number of parameters. Their numerical values can be found in Table B. (i) The DM pattern is scaled by the DM amplitude (microns). (ii) The DM pattern is magnified by a scaling parameter to account for the pupil magnification between the DM and the WFS. (iii) The DM pattern is rotated by an angle ‘azimuth - 12.984° + a reference angle’ that is calibrated on the VLT beacon. (iv) The model does not include the lateral misregistrations because these are corrected in hardware (see Sec. 2.3.5 and Fig. 2). (v) The model includes the two anamorphosis (at 0° and 45°) to capture the pupil stretch. These distortions of the pupil at the coude focus (M9 DIC) and inside the WFS (PBS, WFS optics) can be factorized into a single, static distortion because the DROT/NROT/LROT are kept at a static position.

2.3.5. Lateral registration between DM and WFS

The PSIM model introduced in Sec. 2.3.4 assumes that there is no lateral error between the DM and the WFS. To keep the system closed to its optimal functioning point,

Parameter	LGS	NGS VIS HO	NGS VIS LO	NGS IR HO	NGS IR LO
camera	OCAM2	OCAM2		Saphira	
nb. of pixel read	240×240	240×240		72×72	
nb. of subap	30×30	40×40	4×4	9×9	
plate scale ["/pix]	0.71	0.42	0.21	0.51	
frame rate [Hz]	1000	1000,500,250,100	500,250,100	1000,500,200,100	500,200,100
nb. of controlled KL modes	500	500,200	3	50	3

Table 1. Parameters of the WFS.

any alignment drift is monitored and corrected in hardware by an OS secondary loop, see App. C.

As discussed in Appendix A.1, the photometric pupil and the DM pupil are different and thus the former cannot be used to track the lateral misregistration of the latter. To do so, GPAO implements an innovative approach that is only based on the ‘2D+ t ’ representation of DM command telemetry (Berdeu et al. 2024). In short, in the presence of a lateral misalignment, the symmetric (cosine) part a given spatial frequency of the command is seen with an anti-symmetric (sine) small component by the WFS that leaks on the anti-symmetric part of the DM command, and vice-versa. For small shifts, this coupling pattern only depends on the loop parameters (gain, latency, frequency, ...) and is proportional to the lateral error. Contrary to other methods, this solution is non-invasive nor model-dependent (Heritier et al. 2021) while the sparsity of the signal in the ‘2D+ t ’ Fourier domain makes it fast and with a limited bias by the wind (Béchet et al. 2012; Heritier 2019).

This method has been validated on-sky during the GPAO commissioning, both with the 9×9 CIAO IR WFS and with the 40×40 NGS HO visible WFS. Nonetheless, in strong wind conditions, it still proved to be biased by the wind in the HO mode at low frequencies. As a consequence, the auto-alignment secondary loop is disabled, without noticeable loss of performances, for loop frequencies smaller than 250Hz where only 200 modes are controlled and for which a fine alignment is not necessary.

2.3.6. Photometric pupil stability

Because the DM is actively registered into the WFS (see Sec. 2.3.5), the photometric pupil (M2) is not necessarily centered. The mitigation implemented in GPAO is to zero the slopes measured on poorly illuminated subapertures. Practically, the median flux over the subapertures is measured during the acquisition phase, and a threshold of 15% of this value is defined in the RTC. We verified that the loop remains stable even when vignetting a large fraction of the pupil by rotating the dome. This mechanism is very effective to deal with the offset of the photometric pupil and its wobbling with the elevation axis, which amounted to nearly 3 sub-apertures (in 40×40 mode) on the worst telescope when installing GPAO. Still, the drawback is that a part of the wavefront, opposite to the under-illuminated subapertures, remains uncontrolled because it falls out of the DM.

Considering that an offset of 10% of the pupil diameter was unacceptable, ESO and the consortium quickly engaged into a physical realignment of the four UT telescopes. The M3 mirror was tilted to reduce the elevation wobble. The

M4 mirror, closed to a focus, was adjusted to center the now stable pupil into the DM. Altogether, the photometric pupil is now always within 1 sub-aperture (in 40×40 mode).

2.3.7. Chopping

GPAO implements chopping based on asynchronous communications and absolute time stamping. The ISS receives the chopping parameters (amplitude and direction on sky, period, duty cycle and start time few seconds in the future) from the scientific instrument. The ISS forward these parameters to the M2 controller and to the GPAO OS. The GPAO OS forwards these parameters to the HRTC. Once the start time is reached, the HRTC determines the current state ("on-sky" or "on-science") of the current frame purely based on the period and the absolute time. There is no other synchronization between the M2 and the RTC rather than sharing the same time server. The RTC freezes the loop during the "on-sky" phases, and resumes the loop during the "on-science" phases. Freezing in the RTC means that the last command is kept unmodified, without feedback from the measurements, nor accumulation of leaks or anti-windup. The SRTC monitors and recorders ignore the "on-sky" phases when relevant (e.g to compute the of-floads).

2.3.8. Non-common path aberration

Non-common path aberrations (NCPA) inside the WFS and inside the VLTI can be factorized into a single, static aberration because the DROT/NROT/LROT are kept at a static position. These NCPA are measured all the way to the IRIS tip-tilt sensor of VLTI (Gitton et al. 2004), using the telescope Nasmyth beacon in day time. Sequentially on each controlled mode starting with defocus, a modulation is injected into the closed AO loop at the level of the WFS slope measurements. The flux of the brightest pixel is recorded on IRIS in closed loop. The modulation on each mode contains a high 25 Hz frequency and two periods of a low 0.25 Hz frequency. The amplitude of the 25 Hz detected on IRIS reaches a minimum when the slow period crosses the NCPA offset (the strehl maximum). Two periods are recorded, giving at least three consecutive minima and an easy measurement mechanism from their temporal spacing.

The previous calibration is no longer valid when the beam crosses the M9 and PBS dichroic at a different location, that is when the NGS WFS is off-axis. We model this field-dependent NCPA by a low order polynomial dependence of the astigmatism modes. This model is calibrated on measurements made on the beacon at various off-axis positions, thanks to the steering capability of the QSM.

3. Operations and performances

3.1. Operations

3.1.1. Operational Modes

GPAO supports the four modes described below. Seeing-enhancer and seeing-limited modes are virtually useless for the spatially filtered instruments of VLTI, and are thus not supported. The mode is explicitly requested by the user when designing its Observing Block. It is forwarded to GPAO through the ISS. Changing GPAO mode requires to reconfigure some hardware (closing/opening shutters, moving HO/LO stage, turning filter wheels), but also to fully shut down and restart all the individual SRTC and HRTC processes. This action takes about 200 seconds and actually dominates the overhead when changing modes.

GPAO_NGS_VIS In this mode, the NGS WFS is configured in HO (40×40) and used in stand-alone at up to 1000 Hz. The RTC processes the pixels from the NGS WFS only. It controls 500 KL modes of the DM and offloads the tip/tilt to the telescope axes. The framerate of the HO pipeline as well as the number of controlled modes are lowered when approaching the limiting magnitude.

GPAO_LGS_VIS In this mode, the NGS WFS is configured in LO (4×4) and used as fast tip/tilt/focus sensor up to 500 Hz. The LGS WFS senses the higher modes at 1000 Hz. The RTC processes the pixels from the NGS WFS and LGS WFS. It controls the DM and offloads the tip/tilt to the telescope axis. It also controls the LSM and JM of the Laser projector. It offloads the focus measured in the LGS WFS into the LFOC focalisation device. The framerate of the LO pipeline is lowered when approaching the limiting magnitude.

GPAO_NGS_IR In this mode, the CIAO WFS senses the first 60 KL modes in stand-alone at 1000 Hz. The RTC processes the pixels from the CIAO WFS. It controls 60 KL modes of the DM and offloads the tip/tilt to the telescope axes. The framerate of the pipeline is lowered when approaching the limiting magnitude.

GPAO_LGS_IR In this mode, the CIAO WFS senses the tip/tilt/focus modes at 500 Hz while the remaining ones are sensed with the LGS WFS at 1 kHz. The RTC processes the pixels from the CIAO WFS and from the LGS WFS. It controls the DM, M2, LSM and JM of the LGS projector. The framerate of the IR LO pipeline is lowered when approaching the limiting magnitude.

3.1.2. Automated Acquisition

A VLTI preset (change of target) is triggered by the Observing Block being executed on the instrument (GRAVITY, MATISSE, PIONIER). The preset is sent to the ISS, which forwards it to the various subsystems of the VLTI: telescope, delay lines, STS, laboratory configuration, and GPAO. GPAO immediately initiates its (re)configuration and setup, in parallel to the telescope slewing. Relevant to GPAO, the preset contains the desired adaptive optics mode (see Sec. 3.1.1), the G-magnitude of the natural AO

guide star, the coordinates of the AO guide star, the desired coordinate for the LGS guide star, and the on-axis coordinates of the telescope. The difference between these coordinates drives the position of the NXYT, the LXYT and of the FSM, and therefore the off-axis offset of the wavefront sensors at the coude focus. The NGS wavefront sensor is setup to a low amplification, and to a rate adapted to detect the given G magnitude. A filter is possibly inserted to reduce the flux on very bright objects. If requested, the LGS wavefront sensor is setup to a static configuration (1000 Hz, amplification gain 300). The LGS system is also configured and the laser is shined as soon as the telescope has reached the coordinate and follows a blind trajectory.

The baseline operational scenario of GPAO is to execute the acquisition fully automatically. In short, the acquisition procedure can be decomposed into three individual steps: detecting the star (DETCGS), fine centering and optimization (OPTCGS), and closing the loop (STRTCAG). In LGS modes, these three steps are first applied to the laser guide star, and only then to the natural guide star. The DETCGS executes a raster with the QSM (for the NGS) or with the JM (for the LGS) and moves the telescope (for the NGS) or the LSM (for the LGS) to the position of maximum of flux. Then, the OPCGS estimates the mean position of the star in the subapertures and refines the centering. The DM/WFS registration is optimized following the strategy presented in Berdeu et al. (2024). It consists of modulating a subset of KL modes on the DM, and fitting the measured interaction matrix with a fast correlation method. The actual magnitude of the star is estimated, based on a calibrated zero-point. For the NGS VIS WFS only, this magnitude serves as a starting point for an optimization of the camera amplification gain and frame rate. The last step of the optimization estimates the median flux per subaperture, and updates the threshold of the RTC below which the gradients are considered invalid and forced to zero. At this point, the STRTCAG is executed and all loops related to this WFS are closed, including offloads and secondary loops. Once all required wavefront sensor(s) are acquired, GPAO raises to state "Coudé Guiding", a sound is played, and the ISS informs the scientific instrument to pursue its own acquisition.

The automatic acquisition works well when there are no other bright targets around the guide star. For more complex fields, the operator has the possibility to manually execute the acquisition steps one by one (DETCGS, OPTCGS, STRTCAG) to specifically select the correct guide star.

3.1.3. Coordination with science observations

A typical observation block at the VLTI last for 1h. It includes several exposures of about 6 min (files produced by the OS). Each exposure is composed of several DIT (detector integration time) from 200 ms to 300 s. The GPAO OS is informed via the ISS of the beginning and end of every exposure. There is a typical overhead of few seconds between exposures, coming from operations in the instrument itself.

Telemetry The OS triggers the recording of 30s of AO telemetry at the start of each science exposure. This telemetry includes all frames for the "loop data" (slopes, intensities, commands, chopping state) and a sub-sampling at 10 Hz for the "pixel data" (pixel stream). The files are

stored for latter use, but they are unfortunately not available through the ESO main archive system. At the end of each exposure, the OS collects and forwards the necessary information to build the AO-related part of FITS header.

Passive Support The Unit Telescopes incorporate a mechanism (M1PMOVE) to regularly re-center laterally the M1 mirror itself inside its cell. This is especially important when observing at low elevation. When triggered, the lateral motion of M1 creates a large step of optical path delay, which makes the fringe-tracker lose the tracking for several seconds. Therefore the automatic, regular M1PMOVE mechanism is deactivated when the UTs are used for VLTI. Instead, GPAO requests the M1PMOVE only when in-between the exposures of the instrument. It prevents from losing the fringes inside the coherent integration in the scientific detector. It also hides the time to recover the fringes inside the overhead between the exposures.

3.1.4. Maintenance and monitoring

Calibrations The calibration unit of the WFS can be used during day-time in order to perform internal re-calibrations of the LGS and NGS WFSs: internal alignment, pixelscale, reference slopes, and conversion gain and amplification gain of the OCAM2 camera. No such tool exist for the IR CIAO WFS. The darks reference files of all cameras can be re-calibrated by closing the corresponding shutters. The PSIM parameters (see Sec. 2.3.4) can be re-calibrated during day-time by measuring an interaction matrix using the Nasmyth beacon. An extended version of this re-calibration include measurements at various telescope Azimuth angle, to verify the model pertinence and assess the accuracy of the measurement. The DM reference vector used to flatten the DM in open-loop can be re-calibrated using the Nasmyth beacon. The template first closes the loop with 500 KL modes, then removes all leaks from the system, then inverts a zonal interaction matrix keeping 1100 eigenvectors to create a zonal control matrix, then closes the loop for few seconds and records the DM position.

Daily health check The daily health check template includes a measurement of the gain of all actuators of the DM, to verify functionality. The mean gain is trended over time as well as the number of non-functioning actuators inside the clear aperture (one in GPAO2). The template also includes a measurement of the DM flat reference vector. It is used for trending only and the actual vector installed in the system is not updated automatically.

3.2. Verifications

3.2.1. Transmission

A fit of Fig. D.1 in Appendix D gives the following zero points for GPAO 1 to 4: 1.12, 1.32, 1.32, 1.01×10^{11} e/s for a magnitude 0 in the visible. The ratio between the best and the worst GPAO is 0.75. For the 40×40 WFS running at 100 Hz, the available flux is ≈ 5 electrons per frame and per subaperture for a magnitude 13. This level of illumination corresponds to the operational limiting magnitude.

The zero-point of Gaia Grp (24.7619, see Riello et al. 2021) amount for 5.0×10^{11} e/s when scaled to the collecting

area of the Unit Telescopes (14.74 m^2) with respect to Gaia (0.723 m^2). The chromatic shape of GPAO matches well the one of the Gaia Grp band-pass. The absolute peak transmission of Gaia Grp being 74%, the absolute peak transmission of the visible wavefront sensors of GPAO are in the range 16.5% to 19.1%.

In the infrared, a fit of Fig. D.2 gives the following zero points for GPAO 1 to 4: 1.62, 2.06, 1.51, 1.42×10^{10} e/s for a magnitude 0. When running at 100 Hz, the illumination of 5 electrons per frame and per subaperture is reached at a magnitude 11, which again corresponds to the operational limiting magnitude.

3.2.2. Transfer function

Measurements and model of the transfer function is described in Appendix E. It includes the delay introduced by the integration time, the hold, and the mechanical response of the DM. It also includes an additional pure-delay term T_d to account for the transferring, buffering and computations of data. We adjust this model to the measured rejection functions. Representative results are shown in Fig. E.1. The rejection bandwidth at -3 dB is estimated at 45 Hz in the visible and 50 Hz in the infrared, comparable to those of SPHERE or other extreme adaptive optics. The fitted integration gain K_i are slightly smaller than the value actually used in the controller (0.5). There are several possible explanations such as optical gains or overestimated DM mechanical gains in our geometrical model.

In the visible wavefront sensor, the expected pure-delay $T_d \approx 830 \mu\text{s}$ is the sum of the readout time of the OCAM2 ($500 \mu\text{s}$, equal to the maximum frame rate), the time to extract the pixel from the camera ($47 \mu\text{s}$, from manufacturer), the buffering in the frame-grabber ($110 \mu\text{s}$, frame sent in 5 packets) the HRTC latency ($80 \mu\text{s}$, measured at the oscilloscope), the time to transfer the command to the DM at 1 GbE ($21 \mu\text{s}$), and the buffering inside the DM electronic ($70 \mu\text{s}$, from manufacturer). This prediction matches adequately the measured pure-delay of $T_d = 856 \mu\text{s}$, therefore validating our understanding of the dynamic of the system. A *total loop delay* is often computed by adding the pure-delay (above), half the DM settling time ($400 \mu\text{s}$), and 1 full frame to account for the digitalized controller. This total loop delay is 2.4 frames at 1 kHz frame-rate.

The CIAO detector uses a reset and double-correlated read sequencer. It reduces the duty cycle, but also reduces the pure-delay with respect to our basic model of Appendix E. There is no buffering in a frame grabber. Finally, the computation time is shorter because there is less pixels and slopes to process.

3.2.3. Piston control

Willez et al. (2019b) showed that it is possible to verify the piston-content of the modal basis. A piston conversion factor for each mode is measured by injecting a modulation on the DM and detecting it with the GRAVITY fringe tracker. The conversion factor is the ratio between the piston measured in $\mu\text{m-rms}$ and the modal modulation in $\mu\text{m-rms}$. The modal basis of GPAO is defined to be piston-free when integrating the wavefront with a uniform weight over the telescope pupil. When accounting for the single-mode apodisation of the pupil by the GRAVITY fiber, the model

predicts significant conversion factors for focus and spherical modes (and all radial-only higher orders). We measure a conversion factor of 0.57 for the focus (0.35 expected) and 0.07 for the first spherical (0.05 expected).

3.2.4. Acquisition time budget

A successful automatic acquisition in NGS mode executes the following sequence of OS commands: SETUP, DETCGS, OPTCGS, STRTCAG (see Sec. 3.1.2). Acquisition duration is measured from the start of DETCGS triggered after one cycle of active optics corrections to the end of the STRTCAG. Therefore, all setup activities that are parallelized with the telescope slew are not accounted for. Results give a median acquisition duration of 110 seconds, with 95% of the acquisitions lasting less than 180s. The LGS acquisition is typically 30 seconds longer than the NGS.

3.3. System-level performances

3.3.1. Strehl vs. magnitude and seeing

Figure 3 presents the measurements of Strehl collected during the commissioning nights versus seeing and magnitude of the guide star for the NGS VIS mode. The values closely match the prediction based on scaling from ERS.

The overall performances are significantly improved with respect to the former MACAO but the limiting magnitude is about the same. This is expected since the faint-end will be covered by the LGS mode once available. A decisive advantage of GPAO is its improved robustness against degraded seeing conditions. We estimate that it increases the operational time of VLTI by at least 30%.

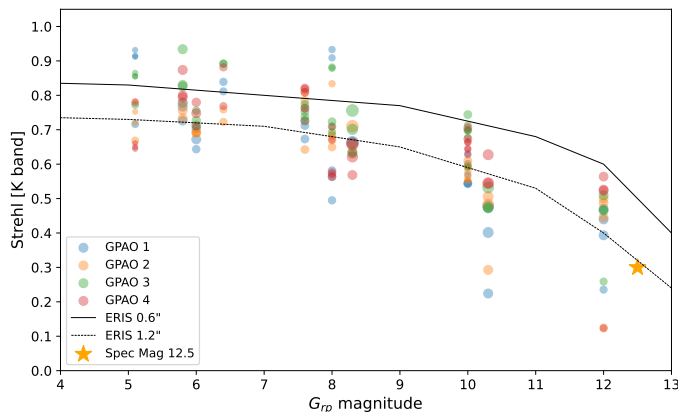


Fig. 3. Strehl in the K-band versus the Grp magnitude of the guide star. The seeing conditions are proportional to the diameter

3.3.2. Injection stability

Interferometric fringe tracking is particularly sensitive to occasional flux dropouts, which often make the fringe tracker lose the white-light position, and lead to a long recovery time before a truly coherent integration can resume. Tatulli et al. (2010) theorized the effect of improved adaptive optics in the coupling into a single mode fiber, with a specific emphasis on the stability of the coupling for optical interferometry. We quantified the improvement brought

by GPAO by using the same metric as used for NAOMI in Woillez et al. (2019b). This metric, noted $P_{5\%}/P_{95\%}$, represents the ratio of the flux at low 5% injection to the flux at high 95% injection. It tends to one for a perfectly stable injection, and tends to zero in the presence of regular flux dropouts. Figure 4 compares the histogram of the $P_{5\%}/P_{95\%}$ metric for GPAO (blue) and MACAO (orange) for each of the UTs. The measurements are from the GRAVITY Fringe Tracking instrument, in the K-band. The clear shift towards higher $P_{5\%}/P_{95\%}$ demonstrates a significant improvement in the stability of the flux injected into the single mode fiber of the fringe tracker. Median values for MACAO are 0.38, 0.33, 0.25, and 0.39 for UT4, UT3, UT2, and UT1 respectively. Median values for GPAO are 0.58, 0.64, 0.62, and 0.69 indicating a factor of 1.5-3 improvement.

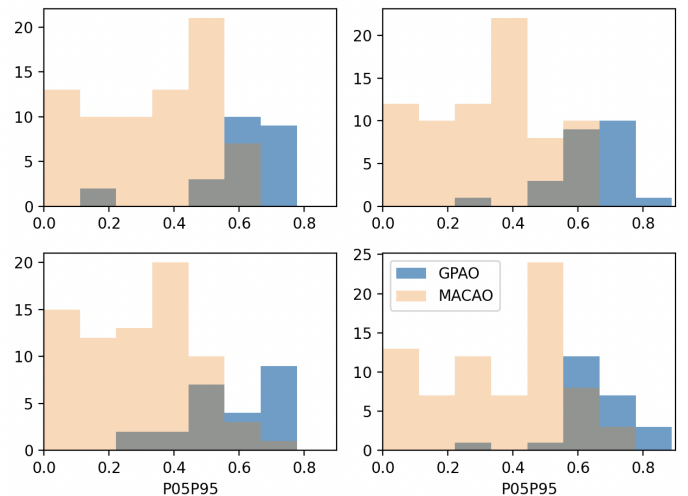


Fig. 4. Comparison between MACAO and GPAO AO of the $P_{5\%}/P_{95\%}$ injection metric histograms for the GRAVITY fringe tracker. The flux dropouts, represented by $P_{5\%}/P_{95\%}$ values close to zero, are significantly reduced by GPAO.

3.3.3. Fringe tracking performances

Figure 5 compares the optical path delay residuals as a function of the SNR of the Fringe Tracker for GPAO and MACAO, for two independent baselines. The faint end displays the expected dependency between the SNR and the performance for a noise-limited system. The bright end shows the expected plateau of a rejection-limited system. The improvement brought by GPAO is obvious, with residuals just above 100 nm compared to 150 nm when the fringe tracker was fed by MACAO.

This positive outcome of the installation of GPAO was not anticipated. The best explanation comes from the change in deformable mirror technology. The MACAO curvature deformable mirror, with influence functions highly coupled to the piston mode, was replaced by the ALPAO deformable mirror, with very localized influence functions. We believe that the MACAO system was somehow coupling the high spatial and temporal frequency atmospheric perturbations back into a high temporal frequency piston.

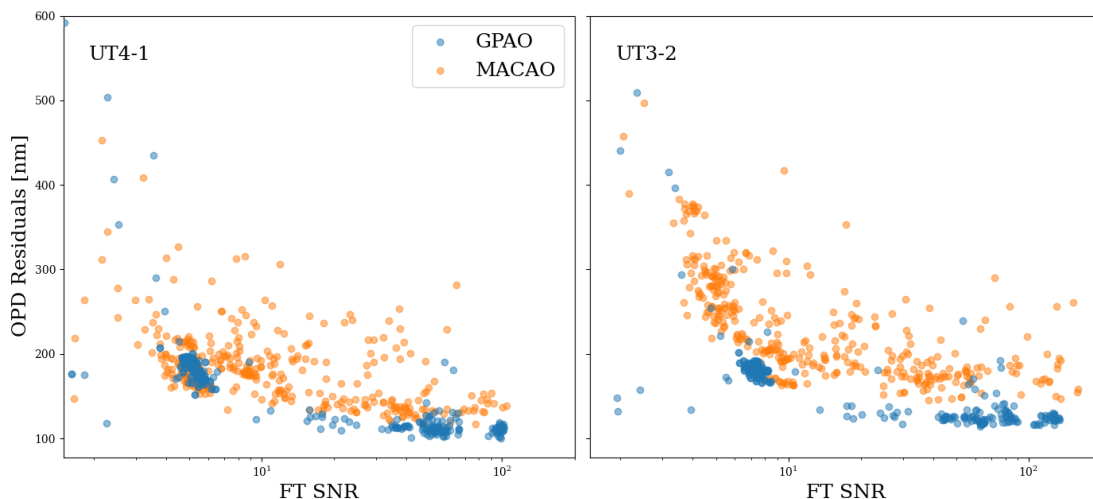


Fig. 5. Comparison between MACAO and GPAO AO of the Optical Path Delay residual after fringe tracking, as a function of the SNR in the Fringe Tracker. Data are shown for two independent baselines, representative of the performance of the entire array.

4. Enabling new science

This section presents the observing modes enabled by GPAO. We illustrate this new range of capabilities with several archetypical objects which were acquired during Fall 2024 commissioning.

4.1. High-redshift quasar $z=3.9$

Extragalactic science depends critically on sensitivity. Within GRAVITY+, the first part of the infrastructure upgrade was the implementation of a wide field-of-view capability, allowing a separation of up to 30 arcsec between the FT and the SC channel (GRAVITY+ Collaboration et al. 2022) and enabling the first dynamical mass measurement of a $z \sim 2$ Super Massive Black Hole (SMBH) (Abuter et al. 2024). To push even further into the early Universe as well as allow for building statistical samples to probe SMBH-galaxy coevolution across cosmic time requires an increase in both sensitivity and sky coverage for GRAVITY, because of the need for off-axis reference stars to perform adaptive optics and fringe tracking. The new GPAO systems with high-order corrections achieve both through the improved the Strehl and fiber injection efficiency which increases the SNR on the interferometric observables and pushes the fringe tracking limiting magnitude to fainter objects.

Expanding the capabilities of GRAVITY to dynamically measure black hole masses at earlier times and fainter objects is also very timely given recent results from the James Webb Space Telescope (JWST). JWST-discovered broad-line active galactic nuclei (AGN) at $z > 4$ suggest a fast growth of black holes that outpaces their host galaxy and leads to overmassive black holes (e.g. Maiolino et al. 2024). These masses are derived from local scaling relations extrapolated to high redshift and indeed significant changes in the gas structure can lead to up to two orders of magnitude difference in the inferred mass (e.g. Maiolino et al. 2025; Naidu et al. 2025). Direct measurements of the mass are therefore critical in our understanding of black hole growth.

Here, we illustrate the observations of a redshift $z \sim 4$ quasar with GPAO-NGS in combination with the Gravity-

Wide mode. We selected SMSS J052915.80-435152.0 ($m_K = 13.9$), hereafter called J0529-4351, which was discovered to be “the most luminous quasar in the Universe” (Wolf et al. 2024). Its high luminosity would predict a very large BLR differential phase signal and its redshift places both the $H\beta$ and $H\gamma$ recombination lines in the K -band. J0529-4351 is also located 22 arcsec away from a star with $G_{RP} = 12.1$ and $m_K = 11.0$ (see Fig. 6 left), making it a good test of the capabilities of GPAO-NGS at the faint end as well as the new FT limiting magnitude.

We observed J0529-4351 with the MEDIUM spectral resolution grism and COMBINED polarisation mode over five nights. The atmospheric conditions ranged from $0.4''$ to $1.1''$, and required overall good atmosphere conditions given the large separation between the AO/FT star and the quasar, which affects the anisoplanetic errors. The total usable data is about 7.8h on-target, for which the improved performance of the AO around seeing $\geq 0.5''$ is crucial, and previously not possible with MACAO. The raw data with $\text{DIT}=100\text{s}$ and $\text{NDIT}=4$ were reduced through the GRAVITY pipeline to produce complex visibilities and differential phases. The differential phase data were analyzed following the methodology described in Abuter et al. (2024). In the middle panel of Fig. 6, we show the total photometric flux of J0529-4351 across the $H\beta$ emission line and in the right panel is the average differential phase spectrum by combining the longest three baselines (UT1-UT4, UT1-UT3, and UT2-UT4). The redshift of the quasar places the line near the edge of the wavelength range of GRAVITY and therefore cuts off part of the red wing. Nevertheless, we strongly detect the $H\beta$ line which peaks at $2.41 \mu\text{m}$ and has a FWHM of $\sim 3200 \text{km.s}^{-1}$ with an underlying blue wing offset by $\sim 2000 \text{km.s}^{-1}$ and a FWHM of $\sim 7000 \text{km.s}^{-1}$. We measure a typical noise of $\sim 1^\circ - 2^\circ$ per spectral channel and per baseline within the $H\beta$ line for the total observing time. The observations show a strong dip in differential phase, which demonstrates that we spatially resolve the BLR in J0529-4351. However, the differential phase signal does not follow a canonical “S-shape”, which indicates a motion departing from pure-Keplerian motion. This is a potential signature of significant radial motion in J0529-4351, which has also been reported in GRAVITY observations

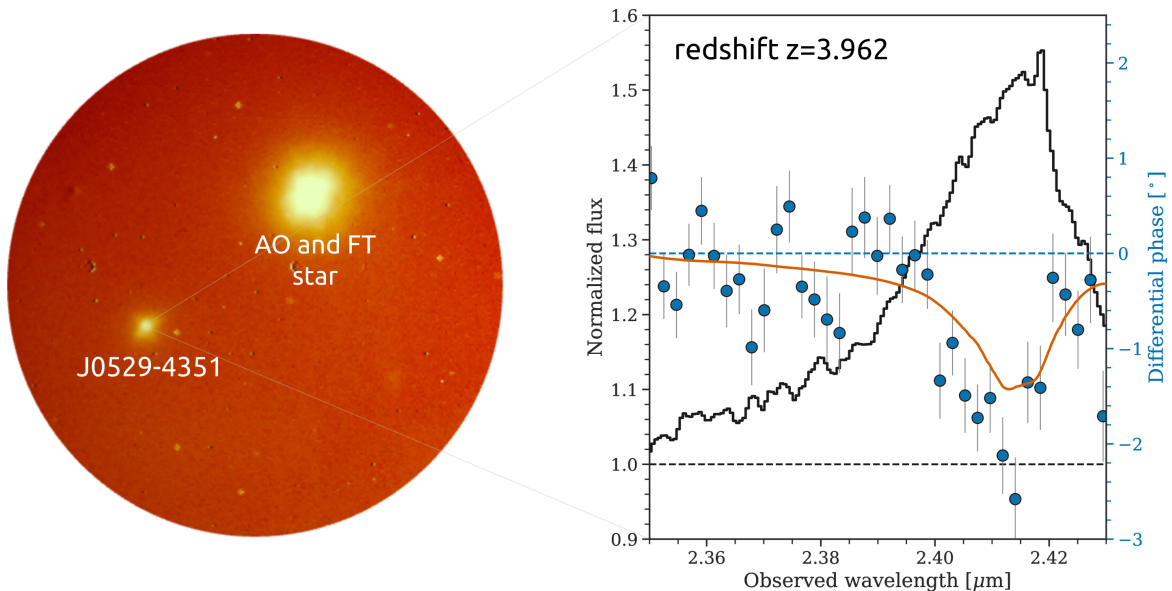


Fig. 6. Left: GRAVITY Acquisition Camera image of the observation of J0529-4351 showing the quasar and AO/FT reference star. On sky the separation is $22''$ but the VLTI optics bring them to GRAVITY in separate beams with a $2''$ separation. **Right:** Averaged differential phase spectrum of J0529-4351 (blue points) across the $H\beta$ emission line (black line). An initial BLR model fit to the line profile and differential phase shows a much smaller BLR and SMBH mass.

of local AGNs GRAVITY Collaboration et al. (2024a), but which has never been observed at high-redshift. The modeling of the BLR in this complex environment would indicate a BLR size and SMBH mass a factor of two smaller than the size predicted from scaling relations, with a measured size from GRAVITY of 1.2 pc , and a SMBH mass of $10^{9.6} M_{\odot}$. We caution the reader these are only preliminary results and a full analysis of the GRAVITY data for this quasar, including the $H\gamma$ line, is beyond the scope of this paper and will be presented in an upcoming dedicated paper.

4.2. Extragalactic Stellar Interferometry

Stellar physics has been a dominant field of infrared interferometry (or "Stellar interferometry"). With the capability to observe faint targets and wide off-axis fringe-tracking targets, it now becomes possible to extend stellar interferometry outside of the Milky Way, in our local Universe. We demonstrate this capability on the Large Magellanic Cloud (LMC), which was observed on 27 September 2024 using GPAO and G-Wide. We focused on the dense central star cluster R136 of the Tarantula Nebula which hosts a large concentration of O and Wolf-Rayet stars within a 5 parsec region (Feast et al. 1960), and are the most massive stars known currently in the Universe (Crowther et al. 2010). These observations allowed to validate the acquisition of targets in crowded fields, using the WFS technical camera for the acquisition of the AO star (see 2.2.2), and the GRAVITY Acquisition Camera in G-Wide mode (GRAVITY+ Collaboration et al. 2022), see Fig. 4.3, upper panel. We observed the star R136c with a magnitude $m_K = 11.31$ with GRAVITY, using the star R136a1 at a separation of 3.35 arcsec for AO ($m_V = 12.8$) and FT ($m_K = 11.15$). The observations were performed in good weather conditions, with seeing $\sim 0.45''$ and wind speed $\sim 7\text{ m/s}$. In only 8 min of exposure on source (DIT=3s) and a spectral resolution $R \simeq 20$, the precision obtained on the closure

phases is of the order of $\sim 1\text{ deg}$ per channel (see Fig. 4.3, lower panel). We analyzed the closure phases to look for a signature of binarity. Despite the short observing time on target, we can rule out binary companions in R136c with a flux ratio of ≥ 0.1 and separation $\gtrsim 5\text{ mas}$. These observations exclude massive companions ($M \gtrsim 20 M_{\odot}$) at large separation $\gtrsim 250\text{ au}$. Significantly stronger constraints on the flux ratio and the separation $100 - 1000\text{ au}$ can be obtained in the future with longer integration and improved uv-coverage. These constraints are very complementary to RV data, which are mainly probing periods shorter than tens of years, corresponding to companions with a separation $\lesssim 100\text{ au}$.

4.3. High-resolution spectroscopy of exoplanets

The upgrade from MACAO to the high-order correction of GPAO is crucial for the observations of exoplanets. This science case benefits both from the improved rejection of the starlight and the better flux injection of the planet in the science fiber. In the following, we illustrate this improvement with high-resolution spectroscopy of exoplanets and brown dwarfs.

4.3.1. Spectroscopy down to $K_{mag} \sim 21$

We observed the ultracool brown dwarf HD 4113 C (Cheetham et al. 2018) with GRAVITY and GPAO during the November 2024 commissioning run, together with the FAINT mode on GRAVITY (Widmann et al. 2022). This substellar companion is located at a separation of $0.7''$ from its host star, with a magnitude about 21 mag in the K band, significantly fainter than the faintest companion successfully observed with the old MACAO system, which was 51 Eri b at about 18.5 mag in the K band at a similar separation. The observations were obtained in average weather conditions with seeing $\sim 0.6 - 1''$. The exposure

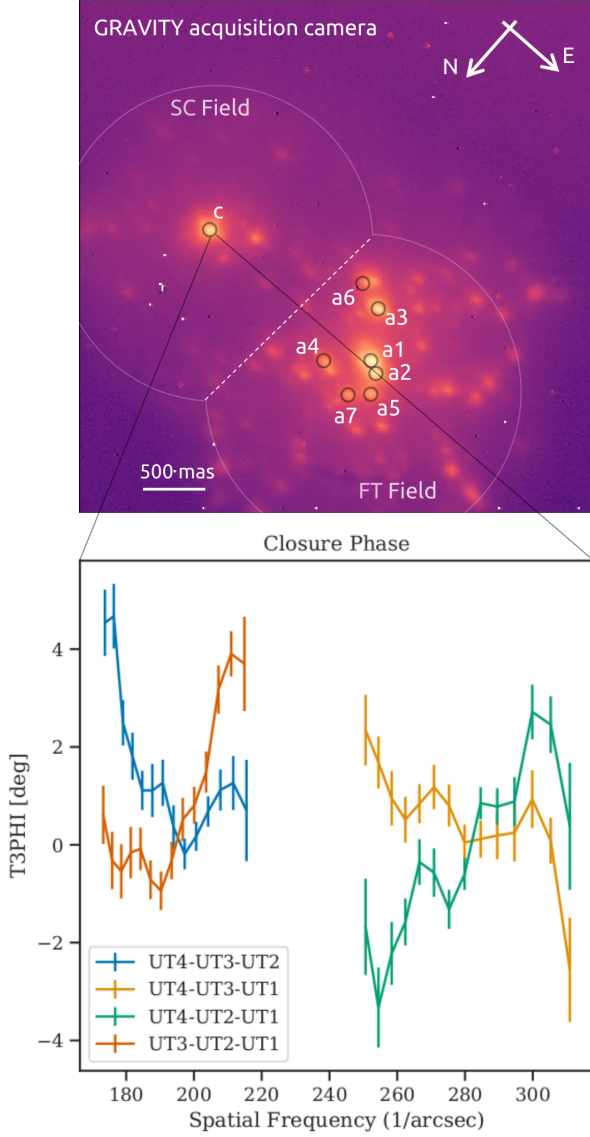


Fig. 7. Upper panel: RMC 136 seen on GRAVITY Acquisition Camera. **Lower panel:** Closure-Phases of RMC136c observed with Gravity-Wide and GPAO. These observations are close to the limiting magnitude in VIS mode, and will largely benefit from the LGS next year.

on HD 4113 C were followed by a swap on a binary calibrator HD 25535 (Nowak et al. 2024a) and reduced using the standard exoGravity tools (GRAVITY Collaboration et al. 2020c). We used long exposure DIT=100s to maximize the SNR. In about 1.7h integration time on-target, we obtain an astrometry accuracy of about $\sim 50 - 100 \mu\text{as}$. We measure the spectrum of HD 4113 C with $R \sim 500$, showing a strong methane absorption band from $2.1 \mu\text{m}$ to $2.45 \mu\text{m}$. The typical sensitivity obtained of HD 4113 C in 1.7h observations is comparable to the sensitivity obtained on 51 Eri in 5.4h observations, as shown in Fig. ??, illustrating the major improvement of Signal-to-Noise-Ratio between MACAO and GPAO. From previous observations, HD 4113 C has an estimated effective temperature $T \sim 500 - 600 \text{ K}$ and a surface gravity $\log g = 5$ (Cheetham et al. 2018). Its effective temperature is in tension with the isochronal mass estimate of $\sim 36 M_{\text{Jup}}$, making this target a potential binary

brown dwarf candidate, as recently demonstrated in G1229 B using GRAVITY resolving power (Xuan et al. 2024).

4.3.2. High-Resolution Spectroscopy of HR8799e

The young giant planet HR 8799 e was observed using GRAVITY and GPAO, which was the first exoplanet detected using optical interferometry (GRAVITY Collaboration et al. 2019b). This object is located at a separation $\sim 0.4''$ of its host star and has a magnitude 15.9 mag in the K band. The observations were carried out during the night of 28 September 2024, with a seeing $\sim 0.7''$. The observing time on target was only 0.3h, but shows the significant improvement in terms of sensitivity compared to previous observations. The new spectrum obtained with GPAO is shown on Fig 9, together with an old HR 8799 e spectrum from 28 August 2021 obtained with MACAO in 0.4h on HR 8799 e, both with a spectral resolution $R \sim 500$. For a comparable exposure time on target, an improvement in the SNR of a factor of ~ 10 can be clearly seen. This improvement is both consistent with an improvement of flux injection in the SC fiber by a factor ~ 3.5 and the reduction of noise by a factor ~ 2.5 . The improvement of flux injection is consistent with the increase in the Strehl ratio for a $G_{\text{RP}} = 5.4$ star from GPAO to MACAO, as shown in Fig 3. The noise reduction originates from a better rejection of stellar photons and longer integration (DIT=100s compared to DIT=30s in 2021), the latter being due to improved fringe-tracking and reduced vibrations, as shown in Section 3.3.3. In only 0.3h, it is possible to clearly identify the CO ro-vibrational lines at $2.3 \mu\text{m}$ in the spectrum of HR 8799 e. With long exposure, this new capability opens up the possibility of high signal-to-noise ratio spectroscopy in order to measure accurate C/O elemental abundances or time-resolved spectroscopy for young exoplanets, such as HR 8799 e or Beta Pic b.

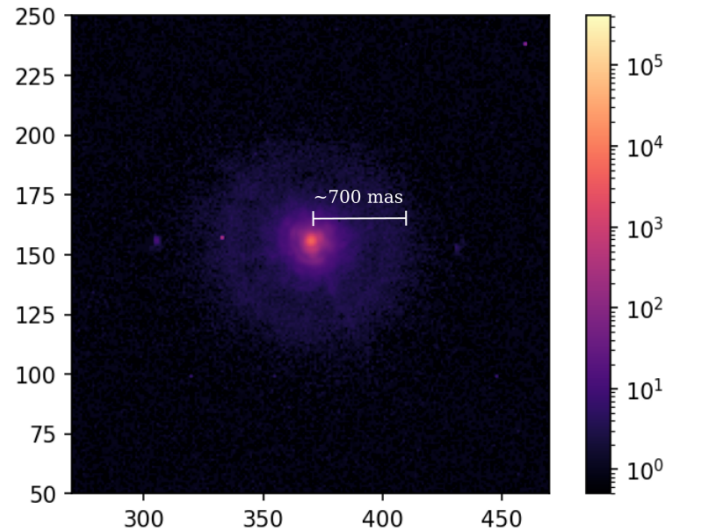


Fig. 8. Image of the GRAVITY Acquisition Camera (H-band) showing the separation of GPAO on HD 4113. The separation to the SC fiber is 700 mas. The ring structure corresponds to the cut-off frequency of the ALPAO DM.

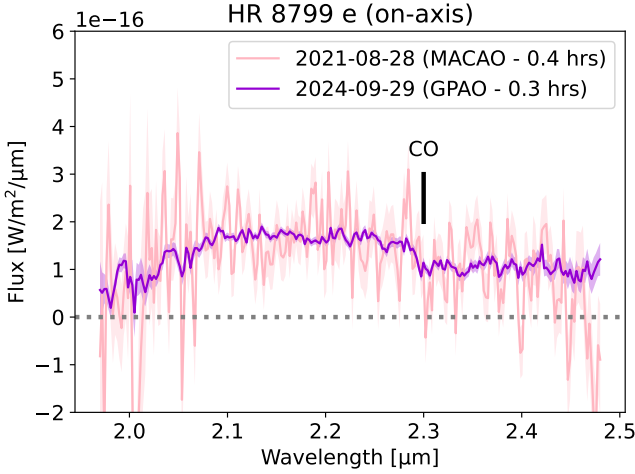


Fig. 9. Two GRAVITY on-axis spectra of the young giant planet HR 8799 e, one obtained with the new GPAO system and one obtained with the old MACAO system.

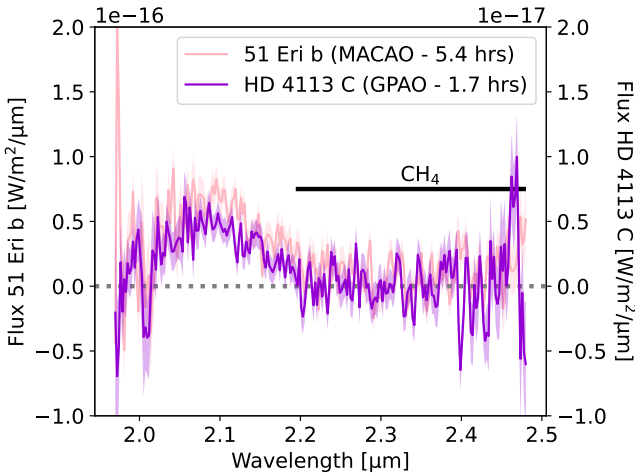


Fig. 10. GRAVITY off-axis spectrum of the brown dwarf companion HD 4113 C obtained with the new GPAO system compared to the off-axis spectrum of the exoplanet 51 Eri b obtained with the old MACAO system. We note the two different y-axes for the two objects, showing that a spectrum with a similar SNR can now be obtained with the new GPAO system for a ~ 10 times fainter object as compared to the old MACAO system.

4.3.3. Mid-infrared L+M spectroscopy

Beta Pic b was observed with GPAO and MATISSE, using the GRA4MAT mode (Woillez et al. 2024). The observations were carried out during the night of 17 November 2024, with typical seeing conditions of $\sim 0.6''$. The MATISSE observations were obtained in L/M bands, using the MEDIUM resolution $R \sim 500$. From the coherent flux, the spectrum of Beta Pic b was extracted, following the methodology described in Houlle et al. (2025). Figure 11 shows a comparison of this new GPAO spectrum with the MACAO spectrum extracted the same way. We find that the MATISSE/MACAO spectrum is scaled by a factor $\approx 4/3$ (+30%) compared to the MATISSE/GPAO one. This means that between the two observations, either the flux of the star is underestimated, or the flux of the planet

is overestimated in MACAO observation, or the other way around for GPAO observation. MATISSE uses spatial filtering, and the GRA4MAT narrow offaxis mode was still in development at the moment of the MACAO observation, with visible drifts in the data. Nevertheless, the comparison of both spectra show the GPAO one has a higher S/N compared to the MACAO one, despite being observed the same amount of time. [TBD: show a full K-L-M spectrum of beta pic b]

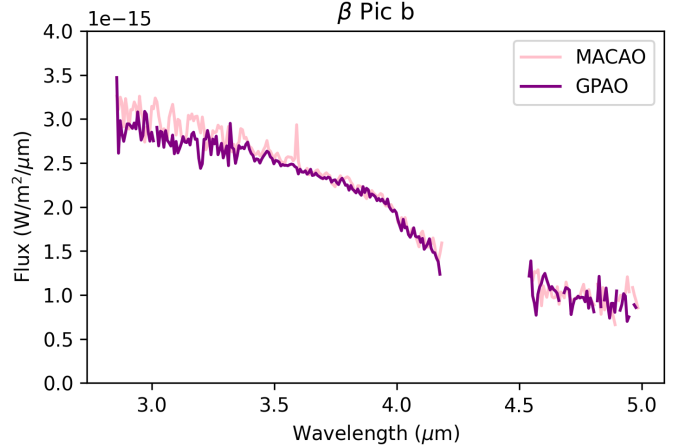


Fig. 11. Comparison of *beta* Pic b spectrum in the L- and M-bands with MATISSE using MACAO (pink) and GPAO (purple) scaled by a factor 4/3 (TBC).

4.4. Embedded Young Stellar Objects

The study of the inner astronomical unit of Young Stellar Objects (YSOs) has advanced significantly thanks to infrared interferometry (Lazareff et al. 2017; GRAVITY Collaboration et al. 2019d; Benisty et al. 2023), but was essentially limited to Class II objects. This limitation originates from the extinction of the envelope surrounding these systems, which are bright in the K band but faint in the G band. The LGS-AO removes this limitation on the color of the AO target, making accessible Class I and T Tauri stars on statistical scale. The NGS-IR mode enables these observations as well, in the specific case where a bright star in the K-band is located in the vicinity of the science target ($\sim 20''$).

We used the NGS-IR to observe the protoplanetary disk in HL Tau. HL Tau is a T Tauri star ($M = 1.7 M_{\odot}$), located in Taurus ($d=147$ pc). This system has been a prime target of sub-millimetric interferometry and ALMA (ALMA Partnership et al. 2015), but was never observed with GRAVITY due to its color ($R - K = 7.0$). The GRAVITY observations were performed by closing the NGS-IR loop on the star XZ Tau, located at a separation of $22.5''$. The star was observed on 14 December 2024, with 30min integration on source and a typical seeing $0.5-0.7''$. The observations were carried out with high-spectral resolution $R \sim 4000$, which also allows to obtain spectro-astrometry of Br- γ .

In the continuum, the visibility amplitudes show that the inner disk of HL Tau is spatially resolved with GRAVITY, with visibility amplitudes smaller than 1 and consistently decreasing for the longer baseline U1-U4, U2-U3. The preliminary fit of the GRAVITY data converges to a typi-

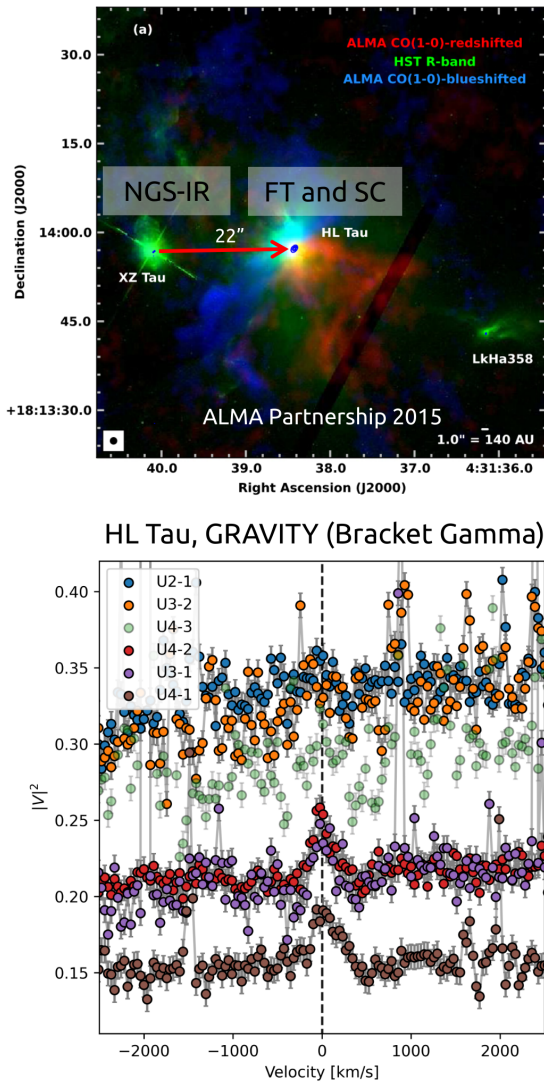


Fig. 12. Upper panel) Overview of the HL Tau region with ALMA and HST, adapted from (ALMA Partnership et al. 2015). The NGS-IR was used on XZ Tau located at 22'' for the AO loop, while fringe-tracking and science was done on HL Tau with GRAVITY. **Lower panel:** The squared visibility of HL Tau, centered around the Br- γ line at 2.16 μm .

cal half-radius of $a \sim 2.0 \text{ mas}$, which is consistent with the values extrapolated from T Tauri stars (GRAVITY Collaboration et al. 2021). The visibility amplitude also includes a significant contribution of an extended flux as probed on the short baseline U1-U2 and U3-U2. This halo corresponds to the envelope in this Class I objects, on GRAVITY Acquisition Camera (not shown here) and on near-infrared and ALMA images (Fig. 12, upper panel). In addition, the differential amplitudes in Br- γ (Fig. 12, lower panel) show that the magnetospheric region in this star is resolved. The differential spectro-astrometry provides a typical precision of $< 1 \text{ deg}$ in HL Tau, which makes it possible to probe the kinematic of Br- γ in the magnetospheric region and the signature of winds. These observations are the first resolved observations of the inner disk of HL Tau below the astronomical scale unit and the first resolved observations of its magnetospheric accretion region. The detailed analysis of

these data is beyond the scope of this paper and will be presented in an upcoming publication.

These observations demonstrate the feasibility to study T Tauri stars and Class I objects with GRAVITY+, and will be greatly generalized with the LGS-AO, which allows to virtually cover all the population of T Tauri stars.

4.5. Sub micro-arcsecond spectro-astrometry

We illustrate the performance on bright targets by analyzing the data recorded on β Pictoris. The star was used as a reference star during an observation of β Pictoris b, for a total integration time on the star of 6 min. Figure 13 shows the spectro-astrometric signal across the photospheric Brackett- γ absorption line. The rotation of the photosphere is very well resolved, with a spectro-astrometric precision of $0.6 \mu\text{as}$ ($1-\sigma$) per spectral channel at $R \approx 4000$.

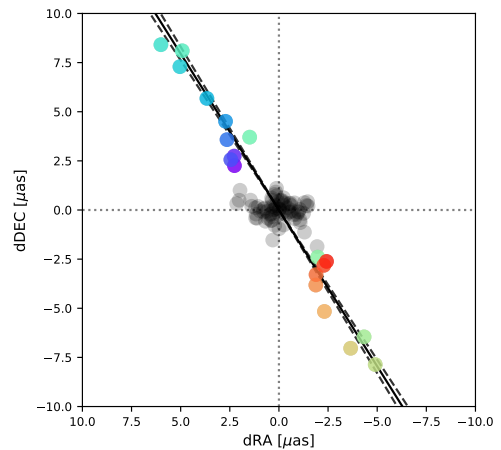


Fig. 13. Spectro-astrometry across the photospheric Brackett- γ absorption line of the star β Pictoris. Blue to red represent velocity channels from -400 to +400 km/s.

5. Conclusions and prospects

GPAO opens up the sensitivity and contrast achievable with VLTI instruments. Our observations demonstrate the performance improvement of GRAVITY with extreme AO correction, achieving fringe-tracking limiting magnitude down to $m_K = 12.5$, combination of high-order correction and wide-field fringe-tracking with microarcsecond spectro-astrometry, spectroscopy on $m_K \sim 21$ substellar objects with $R = 500$, an improvement of a factor 10 in sensitivity on exoplanet observations at typical separations $> 0.5''$, and fringe-tracking residuals $< 100 \text{ nm}$ on the UTs for bright targets.

For future prospects, GPAO-NGS will be the workhorse of VLTI high-contrast mode for exoplanet observations. The first avenue is the implementation of wavefront control and so-called dark-hole techniques (Pouérou et al. 2024). This method consists in injecting modal offsets in the AO to null the starlight at the exact location of the SC fiber. The improvement will benefit particularly short-separation planets $< 60 \text{ mas}$ at 10^{-6} contrast. Even further, the GPAO-NGS will be essential for possible upgrades of VLTI to shorter

wavelengths in the J-band (Lacour et al. 2025), by upgrading the system to 1.5 kHz and 900 corrected modes. Additional prospects include the re-commissioning of secondary guiding between GRAVITY and the Coudé laboratory (Pfuhl et al. 2014) which allows correcting for the VLT tunnel turbulence.

The full deployment of GPAO-LGS with the laser guide stars on all UTs (Gravity+ Collaboration et al. 2022; Booth et al. 2024) will provide the complete implementation of GPAO. The GPAO-LGS mode will offer the capability to perform AO correction up to magnitude $G_{rp} > 18$ with a visible tip-tilt star (LGS-NGS) or with an infrared tip-tilt star (LGS-IR). The LGS mode, combined the wide-field fringe-tracking capability, will increase by two order of magnitude the sky coverage available with GRAVITY. For the AO, the low-order NGS wavefront sensors of GPAO+LGS can control many more modes than tip, tilt and focus (up to 12 modes in the LO VIS sensor and up to 44 modes in the LO IR sensor). The pro (reduced cone-effect) and cons (reduced sensitivity) of increasing this number of modes will be explored in the next years.

The upgrade of GRAVITY and VLT with extreme adaptive optics and the upcoming LGS mode opens up a new parameter space for the study at high angular resolution of the Galactic Center, extragalactic science at high and low redshift, exoplanets at high-contrast and young embedded stars.

6. Summary and outlook

Acknowledgements. We are very grateful to our funding agencies (MPG, ERC, CNRS [PNCG, PNCGRAM], DFG, BMBF, Paris Observatory [CS, PhyFOG], Observatoire des Sciences de l’Univers de Grenoble, Université Grenoble Alpes, Observatoire de la Côte d’Azur, Université Côte d’Azur, Région Sud, and the Fundação para a Ciência e Tecnologia, ANR [contracts AGN Melba ANR-21-CE31-001, ExoVLT ANR-21-CE31-0017]) and the generous support from the Max Planck Foundation - an independent, non-profit organization of private supporters of top research in the Max Planck Society. J-B.LB thanks E. Gendron and Y. Clenet. F.M would like to thank L. Perez. We thank ESO and the ESO/Paranal staff, and the scientific, administrative and technical staff members in our institutions, who helped to make GPAO a reality.

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Appendix A: Missing

- Saturation management

Appendix A.1: Conceptual differences with ERIS

Although GPAO shares a large number of requirements and constraints with ERIS and AOF, there is a set of differences with significant impact on the design choices and development strategy:

- The DM of GPAO does not have an absolute feedback position sensor. The actual DM shape can only be inferred from the command vector sent to the DM.
- In GPAO, the DM (located in M8) and the photometric pupil (located in M2) are two different optical surfaces, linked with a variable wobble and rotation angle (the elevation).
- GPAO has access to the Nasmyth beacon of the Unit Telescope, located before the DM (near M4). It allows closed loop tests and calibration of the AO during daytime. ERIS and the AOF instruments do not have access to this tool: AO calibrations and tests can only be executed at night, on star light.
- GPAO only has a loose requirement regarding the stability of the field sent to the science instrument ($< 0.2''$ for GPAO with respect to $< 0.01''$ for ERIS). Indeed, the VLTI provides a secondary tip/tilt guiding to GPAO, at a rate of 1 Hz, ensuring a constant fine alignment of the field.
- Interferometric fringe tracking requires a diffraction-limited beam, at least partially. The disappearance of a coherent core in the delivered PSF corresponds to the practical limiting magnitude. Therefore, there is no such modes as "seeing enhancer" or "seeing limited" in GPAO.

Appendix B: Parameters of the PSIM model

Appendix C: Secondary Loops

The Secondary Loops run in the OS software, typically at a period of few seconds. The function of most of them has already been introduced in the previous sections. We provide here a simple but exhaustive summary.

SL1 offloads the DM shape in the M1. This loop has been decommissioned because the DM has enough stroke.

SL2 guides the DM registration on the NGS WFS using the NPSM.

SL3 guides the DM registration on the LGS WFS, using the LPSM.

SL4 calculates the NCPA modal vector and reference slopes to be sent to the SRTC.

SL5 updates the rotation angles of the PSIM model in the SRTC, which triggers the recomputation of the Control Matrix. It also updates the rotation angles of various monitoring components.

SL6 guides the DM registration on the CIAO WFS using the VCM of the STS.

SL7 offloads the focus measured in the LGS WFS into the LFOC device.

SL8 offloads the DM mean shape. The offload goes to the telescope axes when observing on-sky or goes to the QSM when using the Nasmyth beacon.

SL9 implements the pointing model and the tracking trajectory for the laser Jitter Mirror, by regularly updating the SRTC, in order to maintain the laser spot inside the LGS WFS even when the Jitter loop is open.

Appendix D: Transmission

Figure D.1 and Figure D.2 show the total number of electrons received on the detector of the visible WFS and of the infrared WFS as a function of the Gaia red-pass (Grp) or K-band magnitude of the guide star. The figures include all science observations from December 2024 to February 2025. We consider a detector conversion gain of 30 e/adu for the OCAM2 camera and 10 e/adu for the Saphira IR detector. The measurements are corrected for the amplification gain.

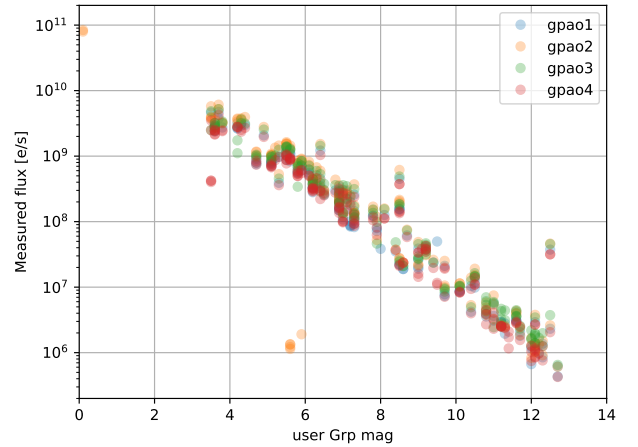


Fig. D.1. Measured total flux on the visible WFS of GPAO versus the Grp Gaia magnitude.

Appendix E: Transfer Function

Measurements (dots) and models (lines) for the wavefront transfer function are presented in Fig. E.1.

The DM is modeled with a sum of 3 second-order resonances with frequencies $s_i = \{1055, 1265, 1600\}$ Hz, damping coefficients $\zeta_i = \{0.025, 0.025, 0.1\}$ and amplitudes $a_i = \{0.33, 0.4, 0.27\}$:

$$DM(s) = \sum_i \frac{a_i}{\left(\frac{s}{s_i}\right)^2 + 2\zeta_i \frac{s}{s_i} + 1} \quad (\text{E.1})$$

Parameter	Description	Unit	GPAO1	GPAO2	GPAO3	GPAO4
COMMON						
DM inclination	design					
DM stroke	measured	μm	3.9	3.7	3.8	3.9
NGS VIS 40×40 (1240 subap)						
DM pitch scaling	measured	%	102.8	103.5	102.2	103.5
DM/WFS registration angle	az-12.984+	deg	0.71	0.71	-1.28	-2.96
WFS anamorphosis 0	measured	%	0	0	0	0
WFS anamorphosis 45	measured	%	0	0	0	0
LGS 30×30 (704 subap)						
DM pitch scaling	measured	%	98.3	98.3	98.3	101.9
DM/WFS registration angle	az-12.984+	deg	-88.43	-88.43	-88.43	-88.43
WFS anamorphosis 0	measured	%	0	0	0	0
WFS anamorphosis 45	measured	%	0	0	0	0
CIAO IR 9×9 (68 subap)						
DM pitch scaling	measured	%	96.4	95.4	101.1	97.7
DM/WFS registration angle	az-12.984+	deg	78.71	80.01	83.81	81.11
WFS anamorphosis 0	measured	%	0.6	3.7	-0.4	0.3
WFS anamorphosis 45	measured	%	0.6	-0.6	-0.8	0.0

Table B.1. Parameters of the PSIM models.

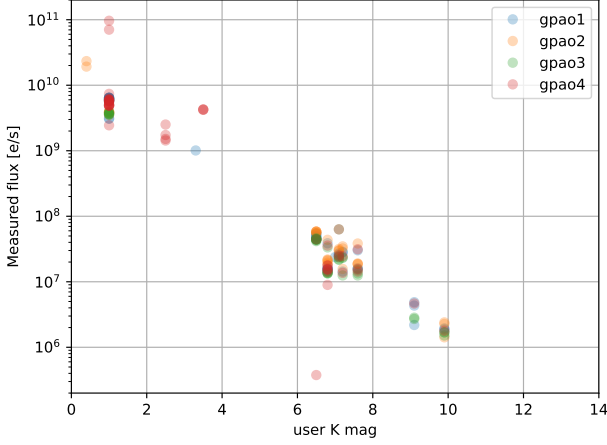


Fig. D.2. Measured total flux on the infrared CIAO WFS versus the K band magnitude.

The WFS is modeled as a discrete integration window of integration time T_i :

$$WFS(s) = \frac{1 - e^{-sT_i}}{sT_i} \quad (\text{E.2})$$

The controller is modeled as a discrete integrator with gain K_I and frame period T_c :

$$C(s) = \frac{K_I}{1 - e^{-sT_c}} \quad (\text{E.3})$$

The output command is hold for one frame period T_c resulting in an additional:

$$H(s) = \frac{1 - e^{-sT_c}}{sT_c} \quad (\text{E.4})$$

All other delays are factorized into a pure-delay T_d :

$$D(s) = e^{-sT_d} \quad (\text{E.5})$$

The closed-loop rejection function writes: $R(s) = 1/(1 + DM(s).WFS(s).C(s).H(s).D(s))$.

Appendix F: Nomenclature

ADC	Atmospheric Dispersion Compensator, made of 2 rotating prisms.
CIAO	Coudé Infrared Adaptive Optics
CM	Control Matrix
CO	Corrective Optics assembly, hosting the QSM and the DM, located in the M8 of the UT
Coudé	Train of mirrors M4 to M8 of the UT, co-rotating with azimuth
DCS	Detector Control Software
DM	Deformable Mirror
DCR	Dichroic splitting the blue (reflected toward the technical camera) and the red (transmitted toward the NGS WFS camera).
ERIS	Enhanced Resolution Imager and Spectrograph
FSM	Field Steering Mirror of the STS, actuates the tip/tilt
GbE	Gigabit Ethernet
GPAO	GRAVITY Plus Adaptive Optics
Guiding	The device follows a closed-loop trajectory (e.g from feedback position sensor)
HSDL	High Speed Data Link
HKL	House Keeping Link
HRTC	Hard Real Time Calculator, performs the real-time tasks
ICS	Instrument Control Software
IM	Interaction Matrix
IF	Influence Functions
JM	Jitter Mirror of the LGS Launch telescope, fast actuator to stabilize the observed position of the LGS spot in the WFS sensor.
KL	Karhunen-Loève modes, in GPAO these modes are defined in DM command space.

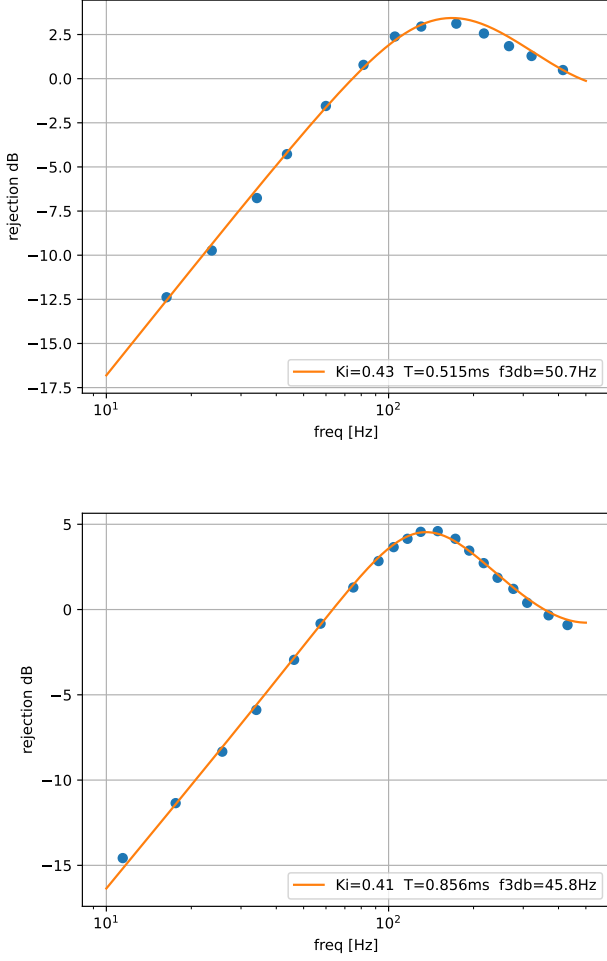


Fig. E.1. Measurements (dots) and models (lines) for the wavefront transfer function for the infrared (top) and visible WFS (bottom), configured at the maximum frame rate of 1000 Hz. The model is fitted to the data by adjusting the integral gain K_i and the pure-delay T_d . The rejection bandwidth defined at -3 dB is also given. Data are presented for GPAO2 but similar results are obtained on the other systems.

- LCU Local Controller Unit, local controller to interface with the hardware, generally based on VXworks in the ESO context
- LFOC Laser Focus Control, long range longitudinal motion of the entire LGS WFS assembly to maintain the focalisation toward the Sodium layer. The range extend toward infinity to allow observing the Nasmyth beacon of the UT.
- LGS Laser Guide Star
- LSM Laser Steering Mirror (called FSM in ERIS/AOF), slow offload actuator for the JM.
- M2 secondary mirror, pupil stop of the UT.
- M9 DIC Dichroic at the coudé focus of the UT, splitting the IR (reflected toward STS) and the VIS (transmitted toward GPAO WFS).

- NFB Nasmyth Focus Beacon, a set of source at the folded Nasmyth focus of the UT, feeding toward the coudé train.
- NGS Natural Guide Star
- OB Observing Block, standardized sequence of observation running on the science instrument.
- OPD Optical Path Difference
- OS Operating Software
- PBS DIC Dichroic at the coudé focus of the UT, splitting the LGS wavelength (reflected toward LGS WFS) and the broad band VIS (transmitted toward GPAO NGS WFS)
- PLC Programmable Logic Controller, local controller to interface with the hardware, generally based on Beckhoff in the ESO context
- PSM Pupil Steering Mirror, lateral pupil actuator (LPSM in the LGS WFS, NPSM in the NGS WFS).
- QSM Quasi Static Mount, gimbal tip/tilt mount of the DM, with slow motion and large steering range.
- RTC Real Time Calculator
- sFPDP Serial Front Panel Data Port
- SL Secondary Loop, supervision, or tracking, or guiding loop executed by the AOO process of the OS. Typical rate 1 to 10s.
- SRTC Soft Real Time Calculator, perform the supervision tasks
- STS Star Separator, located in the VLT coudé room, split the FOV into two region fed to the VLTI
- Tracking The device follow an open-loop trajectory (e.g from coordinates)
- UT Unit Telescope
- VCM Variable Curvature Mirror of the STS, also serving as lateral pupil actuator.
- VLTI Very Large Telescope Interferometer
- VLT Very Large Telescope
- XYT X,Y Table, 2d translation stage to patrol the entire FOV at the coudé focus with the NGS WFS (NXYT) and with the LGS WFS (LXYT)